

SAMSUNG BIOLOGICS Co., Ltd.

Financial Statements

December 31, 2017

(With Independent Auditors' Report Thereon)

Contents

	Page
Independent Auditors' Audit Report	1
Statements of Financial Position	3
Statements of Comprehensive Loss	4
Statements of Changes in Equity	5
Statements of Cash Flows	6
Notes to the Financial Statements	8

Independent Auditors' Report

Based on a report originally issued in Korean

The Board of Directors and Shareholders
Samsung Biologics Co., Ltd.:

We have audited the accompanying financial statements of Samsung Biologics Co., Ltd. (the "Company"), which comprise the statement of financial position as of December 31, 2017, the statements of comprehensive loss, changes in equity and cash flows for the year then ended, and notes, comprising a summary of significant accounting policies and other explanatory information.

Management's Responsibility for the Financial Statements

Management is responsible for the preparation and fair presentation of these financial statements in accordance with Korean International Financial Reporting Standards and for such internal control as management determines is necessary to enable the preparation of financial statements that are free from material misstatement, whether due to fraud or error.

Auditors' Responsibility

Our responsibility is to express an opinion on these financial statements based on our audit. We conducted our audit in accordance with Korean Standards on Auditing. Those standards require that we comply with ethical requirements and plan and perform the audit to obtain reasonable assurance about whether the financial statements are free from material misstatement.

An audit involves performing procedures to obtain audit evidence about the amounts and disclosures in the financial statements. The procedures selected depend on the our judgment, including the assessment of the risks of material misstatement of the financial statements, whether due to fraud or error. In making those risk assessments, we consider internal control relevant to the entity's preparation and fair presentation of the financial statements in order to design audit procedures that are appropriate in the circumstances, but not for the purpose of expressing an opinion on the effectiveness of the entity's internal control. An audit also includes evaluating the appropriateness of accounting policies used and the reasonableness of accounting estimates made by management, as well as evaluating the overall presentation of the financial statements.

We believe that the audit evidence we have obtained is sufficient and appropriate to provide a basis for our audit opinion.

Opinion

In our opinion, the financial statements present fairly, in all material respects, the financial position of the Company as of December 31, 2017 and its financial performance and its cash flows for the year then ended in accordance with Korean International Financial Reporting Standards.

Other Matters

The procedures and practices utilized in the Republic of Korea to audit such financial statements may differ from those generally accepted and applied in other countries.

The accompanying statement of financial position of the Company as of December 31, 2016 and the related statements of comprehensive loss, changes in equity and cash flows for the year then ended, were audited by other auditors in accordance with the Korean Standards on Auditing, whose report thereon dated March 16, 2017, expressed an unqualified opinion.

KPMG Samjong Accounting Corp.
Seoul, Korea
March 14, 2018

This report is effective as of March 14, 2018, the audit report date. Certain subsequent events or circumstances, which may occur between the audit report date and the time of reading this report, could have a material impact on the accompanying financial statements and notes thereto. Accordingly, the readers of the audit report should understand that the above audit report has not been updated to reflect the impact of such subsequent events or circumstances, if any.

Samsung Biologics Co., Ltd.
Statements of Financial Position
As of December 31, 2017 and 2016

(In thousands of won)

	Note	2017	2016
Assets:			
Cash and cash equivalents	6,7,32 ₩	358,088,091	230,337,601
Short-term financial instruments	6,32	-	970,000,000
Trade and other receivables	6,9,30,32	25,817,150	75,630,820
Inventories	10	209,986,717	164,219,980
Deposits provided	6,32	22,605,000	16,289,300
Other current assets	11	4,498,177	4,947,475
Current assets		620,995,135	1,461,425,176
Long-term financial instruments	6,8,32	7,000	8,000
Investments in associates and joint ventures	12,29,30	5,008,234,005	4,944,356,290
Property, plant and equipment	13,30	1,513,332,618	1,090,959,766
Intangible assets	14	18,999,003	14,044,654
Deposits provided	6,32	20,560,000	21,475,000
Other non-current assets	11	963,402	747,220
Non-current assets		6,562,096,028	6,071,590,930
Total assets	₩	7,183,091,163	7,533,016,106
Liabilities:			
Trade and other payables	6,16,19,30,32 ₩	158,214,846	138,056,375
Derivative liabilities	6	1,933,562,427	1,874,470,943
Emission liability	17	502,080	-
Bonds and borrowings	6,15,29,32	138,676,578	403,694,084
Other current liabilities	18	56,486,175	60,323,791
Current liabilities		2,287,442,106	2,476,545,193
Trade and other payables	6,16,19,32	2,650,638	2,300,380
Bonds and borrowings	6,15,29,32	407,235,570	439,363,194
Defined benefit liabilities	19	3,996,064	5,757,265
Deferred tax liabilities	27	505,315,892	526,670,614
Non-current liabilities		919,198,164	974,091,453
Total liabilities		3,206,640,270	3,450,636,646
Equity:			
Share capital	1,20	165,412,500	165,412,500
Share premium	20	2,487,313,082	2,487,313,082
Accumulated other comprehensive income	12,20	(4,009,446)	4,947,005
Retained earnings		1,327,734,757	1,424,706,873
Total equity		3,976,450,893	4,082,379,460
Total liabilities and equity	₩	7,183,091,163	7,533,016,106

See accompanying notes to the financial statements.

Samsung Biologics Co., Ltd.
Statements of Comprehensive Loss
For the years ended December 31, 2017 and 2016

(In thousands of won)

	Note	2017	2016
Revenue	5,21,30 ₩	464,629,334	294,622,021
Cost of revenue	10,22,24,30	(332,829,273)	(268,088,358)
Gross profit		<u>131,800,061</u>	<u>26,533,663</u>
Selling, general and administrative expenses	23,24,30	(65,820,659)	(56,956,302)
Operating profit (loss)		<u>65,979,402</u>	<u>(30,422,639)</u>
Other income	25,30	889,983	1,217,830
Other expenses	25	(136,243)	(39,010)
Finance income	6,26	23,219,775	14,348,567
Finance costs	6,26	(76,345,332)	(86,783,081)
Equity loss on investments in associates and joint ventures	12	(129,602,338)	(129,222,650)
Loss before income tax		<u>(115,994,753)</u>	<u>(230,900,983)</u>
Income tax benefit	27	(19,022,637)	(54,077,573)
Net loss for the period		<u>(96,972,116)</u>	<u>(176,823,410)</u>
Other comprehensive income (loss)		(8,956,451)	107,338
Items that will never be reclassified to profit or loss:			
Defined benefit plan remeasurements	19	(2,189,613)	(1,595,929)
Equity adjustments arising from investments in associates and joint ventures	12,27	682,446	(1,001,039)
Items that are or may be reclassified to profit or loss:			
Equity adjustments arising from investments in associates and joint ventures	12,27	(7,449,284)	2,704,306
Total comprehensive loss for the period	₩	<u>(105,928,567)</u>	<u>(176,716,072)</u>
Loss per share			
Basic and diluted loss per share (in won)	28 ₩	(1,466)	(3,115)

See accompanying notes to the financial statements

Samsung Biologics Co., Ltd.
Statements of Changes in Equity
For the years ended December 31, 2017 and 2016

(In thousands of won)

	Share capital	Share premium	Accumulated other comprehensive income (loss)	Retained earnings	Total equity
Balance as of January 1, 2016	₩ 137,843,605	1,030,620,605	4,839,666	1,601,530,283	2,774,834,159
Comprehensive income (loss) for the year					
Net loss for the year	-	-	-	(176,823,410)	(176,823,410)
Defined benefit plan remeasurements	-	-	(1,595,929)	-	(1,595,929)
Equity adjustments arising from investments in associates and joint ventures	-	-	1,703,268	-	1,703,268
Total comprehensive income (loss) for the year	-	-	107,339	(176,823,410)	(176,716,071)
Transactions with owners					
Capital increase	27,568,895	1,456,692,477	-	-	1,484,261,372
Balance as of December 31, 2016	165,412,500	2,487,313,082	4,947,005	1,424,706,873	4,082,379,460
Balance as of January 1, 2017	165,412,500	2,487,313,082	4,947,005	1,424,706,873	4,082,379,460
Comprehensive loss for the year					
Net loss for the year	-	-	-	(96,972,116)	(96,972,116)
Defined benefit plan remeasurements	-	-	(2,189,613)	-	(2,189,613)
Equity adjustments arising from investments in associates and joint ventures	-	-	(6,766,838)	-	(6,766,838)
Total comprehensive loss for the year	-	-	(8,956,451)	(96,972,116)	(105,928,567)
Balance as of December 31, 2017	₩ 165,412,500	2,487,313,082	(4,009,446)	1,327,734,757	3,976,450,893

See accompanying notes to the financial statements.

Samsung Biologics Co., Ltd.
Statements of Cash Flows
For the years ended December 31, 2017 and 2016

(In thousands of won)

	2017	2016
Cash flows from operating activities		
Net loss for the period	₩ (96,972,116)	(176,823,410)
Adjustments for:		
Retirement benefits	7,637,599	6,377,010
Depreciation	64,188,480	55,784,426
Amortization of intangible assets	5,737,747	4,365,463
Reversal of loss on valuation of inventories	(2,038,561)	(1,275,392)
Loss on disposal of property, plant and equipment	4,946	1,033
Loss on disposal of intangible assets	39,600	-
Gain on foreign currency translations	(3,512,447)	(2,825,232)
Loss on foreign currency translations	331,259	3,962,749
Gain on foreign currency transactions	(6,830,890)	-
Loss on valuation of derivatives	59,091,484	54,031,894
Cost of emissions	502,080	-
Gain from assets contributed	-	(12,018)
Interest income	(9,949,872)	(3,639,650)
Interest expenses	5,683,983	19,254,373
Equity loss on investments in associates and joint ventures	129,602,338	129,222,650
Income tax benefit	(19,022,637)	(54,077,574)
	<u>231,465,109</u>	<u>211,169,732</u>
Changes in:		
Trade receivables	47,807,078	(28,693,573)
Other receivables	(536,295)	(4,435,409)
Other current assets	1,065,034	(2,502,245)
Inventories	(43,728,176)	(43,767,248)
Deposits provided	(5,400,700)	(12,784,300)
Other non-current assets	(990,145)	(359,172)
Trade payables	13,180,780	1,161,707
Other payables	26,582,183	(7,659,178)
Other current liabilities	(3,837,616)	35,397,017
Other non-current payables	1,059,873	1,373,929
Retirement benefits paid	(428,339)	(205,802)
Transfer from affiliates of defined benefit liabilities	89,100	-
Contributions to plan assets	(13,085,879)	(11,056,300)
	<u>21,776,898</u>	<u>(73,530,574)</u>
Cash generated from operating activities	<u>156,269,891</u>	<u>(39,184,252)</u>
Interest received	10,433,121	821,632
Interest paid	(17,036,196)	(24,361,814)
Income tax refunded	149,511	63,136
Net cash from (used in) operating activities	₩ <u>149,816,327</u>	<u>(62,661,298)</u>

See accompanying notes to the financial statements.

Samsung Biologics Co., Ltd.
Statements of Cash Flows, Continued
For the years ended December 31, 2017 and 2016

(In thousands of won)

	2017	2016
Cash flows from investing activities		
Decrease in short-term financial instruments	₩ 2,270,000,000	-
Decrease in long-term financial instruments	1,000	-
Increase in short-term financial instruments	(1,300,000,000)	(970,000,000)
Acquisition of property, plant and equipment	(494,119,453)	(162,992,350)
Acquisition of intangible assets	(10,682,088)	(5,179,126)
Acquisition of investments in associates and joint ventures	(200,014,156)	(234,653,000)
Net cash from (used in) investing activities	265,185,303	(1,372,824,476)
Cash flows from financing activities		
Proceeds from issue of bonds	49,924,480	169,743,600
Proceeds from short-term borrowings	40,000,000	-
Proceeds from long-term borrowings	90,000,000	28,396,169
Capital increase	-	1,484,261,372
Repayment of bonds	(170,000,000)	-
Repayment of short-term borrowings	-	(10,109,490)
Repayment of current portion of long-term borrowings	(236,562,523)	-
Repayment of long-term borrowings	(60,537,587)	(41,000,000)
Net cash from (used in) financing activities	(287,175,630)	1,631,291,651
Net increase in cash and cash equivalents	127,826,000	195,805,877
Cash and cash equivalents at January 1	230,337,601	34,510,062
Exchange rate effects on foreign currency in cash and cash equivalents	(75,510)	21,662
Cash and cash equivalents at December 31	₩ 358,088,091	230,337,601

See accompanying notes to the financial statements.

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2017 and 2016

1. Description of the Company

Samsung Biologics Co., Ltd. (the "Company") was established on April 22, 2011 under the commercial law of the Republic of Korea and is engaged in Biosimilar manufacturing. The Company's head office is located in 300, Songdobio-daero, Yeonsu-gu, Incheon-si.

The Company was listed on securities market in November 10, 2016. As of December 31, 2017, the Company's share capital is ₩165,413 million.

The shareholders of the Company are as follows:

Shareholder	Shares	Percentage of ownership
Samsung C&T Corporation	28,742,466	43.44%
Samsung Electronics Co., Ltd.	20,836,832	31.49%
Others	16,585,702	25.07%
	66,165,000	100.00%

2. Basis of Accounting

These financial statements have been prepared in accordance with Korean International Financial Reporting Standards ("K-IFRS"), as prescribed in *the Act on External Audits of Corporations in the Republic of Korea*.

The financial statements were authorized for issue by the Board of Directors on February 28, 2018, which will be submitted for approval to the shareholders' meeting to be held on March 22, 2018.

(1) Basis of Measurement

The financial statements have been prepared on the historical cost basis, except for the following material items in the statements of financial position:

- ✓ Derivatives financial instruments measured at fair value
- ✓ Liabilities for defined benefit plans recognized at the net of the total present value of defined benefit obligations less the fair value of plan assets

(2) Functional and Presentation Currency

These financial statements are presented in Korean won, which is the Company's functional currency and the currency of the primary economic environment in which the Company operates.

(3) Use of estimates and judgments

The preparation of the financial statements in conformity with K-IFRS requires management to make judgments, estimates and assumptions that affect the application of accounting policies and the reported amounts of assets, liabilities, income and expenses. Actual results may differ from these estimates.

Estimates and underlying assumptions are reviewed on an ongoing basis. Revisions to accounting estimates are recognized in the period prospectively.

(i) Critical judgments

Information about critical judgments in applying accounting policies that have the most significant effect on the amounts recognized in the financial statements is included in the following note:

Note 12 : consolidation scope - whether the Company has de facto control over an investee.

2. Basis of Preparation, Continued

(3) Use of estimates and judgments, continued

(ii) Assumptions and estimation uncertainties

Information about assumptions and estimation uncertainties that have a significant risk of resulting in a material adjustment within the next financial year are included in the following notes: Note 19 : measurement of defined benefit obligations - key actuarial assumptions; Note 27 : recognition of deferred tax assets - use of tax losses; Note 29 : recognition and measurement of provision and contingencies - key assumptions about likelihood and magnitude of an outflow of resources; and Note 14, Note 32 : impairment test - key assumptions underlying recoverable amounts.

(iii) Measurement of fair values

A number of the Company's accounting policies and disclosures require the measurement of fair values for both financial and non-financial assets and liabilities. The Company has an established control framework with respect to the measurement of fair values. This includes a valuation team that has overall responsibility for overseeing all significant fair value measurements, including Level 3 fair values, and reports directly to the Chief Financial Officer (CFO).

The valuation team regularly reviews significant unobservable inputs and valuation adjustments. If third party information, such as broker quotes or pricing services, is used to measure fair values, then the valuation team assesses the evidence obtained from the third parties to support the conclusion that such valuations meet the requirements of K-IFRS, including the level in the fair value hierarchy in which such valuations should be classified.

Significant valuation issues are reported to the Company's Audit Committee.

When measuring the fair value of an asset or a liability, the Company uses market observable data as far as possible. Fair values are categorized into different levels in a fair value hierarchy based on the inputs used in the valuation techniques as follows.

- ✓ Level 1: quoted prices (unadjusted) in active markets for identical assets or liabilities.
- ✓ Level 2: inputs other than quoted prices included in Level 1 that are observable for the asset or liability, either directly (i.e. as prices) or indirectly (i.e. derived from prices).
- ✓ Level 3: inputs for the asset or liability that are not based on observable market data (unobservable inputs).

If the inputs used to measure the fair value of an asset or a liability fall into different levels of the fair value hierarchy, then the fair value measurement is categorized in its entirety in the same level of the fair value hierarchy as the lowest level input that is significant to the entire measurement. The Company recognizes transfers between levels of the fair value hierarchy at the end of the reporting period during which the change has occurred.

Further information about the assumptions made in measuring fair values is included in following note:

Note 6 – Financial instruments

3. Changes in accounting policies

Except for the application of the first and amendment standards as applied beginning January 1, 2017, accounting policies set out below have been applied consistently to all periods presented in these financial statements.

(1) K-IFRS No. 1007, 'Statement of Cash Flows'

Amendments to K-IFRS No. 1007, 'Statement of Cash Flows' require an entity to disclose fluctuations in financing cash flows, changes in the acquisition or loss of control of subsidiaries or other businesses, exchange rate effect, changes in fair values and other changes in liabilities which are classified as financing activities. Accordingly, the information on changes in liabilities from financing activities is included in Note 15.

(2) K-IFRS No. 1012, 'Income Taxes'

Amendments to K-IFRS No. 1012 clarify that the existence of a deductible temporary difference depends solely on a comparison of the carrying amount of debt instruments measured at fair values and their tax base at the end of the reporting period and is not affected by possible future changes in the carrying amount or expected manner of recovery of the debt instruments. When reviewing the recoverability of deferred tax assets, if there is sufficient evidence that it is probable to recover some part of an entity's assets in excess of the carrying amount, the estimated amount of future taxable income would be included in the estimated future taxable income. In addition, future taxable income estimates are calculated by excluding current taxable temporary differences from calculation of taxable income.

Management believes the impact of the amendments on the financial statements is not significant.

(3) K-IFRS No. 1109, 'Financial Instruments'

The Company early adopted K-IFRS No.1109 'Financial Instruments' from the fiscal year beginning on January 1, 2017, an initial application date. The Company applied K-IFRS No.1109 retrospectively; however, the financial statements for the year ended December 31, 2016 presented for comparative purpose were not restated according to paragraphs 7.2.15 and 7.2.26 in transitional provisions of K-IFRS No.1109.

① Financial assets

(a) Classification

From January 1, 2017, the Company classifies its financial assets into the following categories:

- ✓ Financial assets measured at fair value (either through other comprehensive income or through profit or loss), and
- ✓ Financial assets measured at amortized cost

The classification depends on the Company's business model for managing the financial assets and their contractual cash flow characteristics.

For financial assets measured at fair value, gains and losses will either be recorded in profit or loss or other comprehensive income. For investments in debt instruments, such recognition will depend on the business model in which the investment is held.

The Company is permitted to make an irrevocable election at initial recognition to present in other comprehensive income changes in the value of any investment in equity instruments that is not held for trading. Otherwise, any change in fair value of an equity instrument is recognized in profit or loss.

The Company reclassifies debt instruments only when its business model for managing those assets changes.

3. Changes in accounting policies, Continued

(3) K-IFRS No. 1109, 'Financial Instruments', continued

① Financial assets, continued

(b) Measurement

At initial recognition, the Company measures a financial asset at its fair value plus, in the case of a financial asset not at fair value through profit or loss, transaction costs that are directly attributable to the acquisition of the financial asset. Transaction costs of financial assets measured at fair value through profit or loss are expensed in profit or loss.

K-IFRS No. 1109 includes a new classification and measurement of financial assets that reflect the business model for managing the financial assets and their contractual cash flow characteristics.

Financial assets with embedded derivatives are considered in their entirety when determining whether their cash flows are solely payment of principal and interest.

(i) Debt Instruments

Subsequent measurement of debt instruments depends on the Company's business model for managing the financial asset and its contractual cash flow characteristics. The Company classifies its debt instruments into one of the following three measurement categories:

- ✓ Amortized cost: Assets that are held for collection of contractual cash flows where those cash flows represent solely payments of principal and interest are measured at amortized cost. A gain or loss on a debt investment that is subsequently measured at amortized cost and is not part of a hedging relationship is recognized in profit or loss when the asset is derecognized or impaired. Interest income from these financial assets is included in 'finance income' using the effective interest rate method.
- ✓ Fair value through other comprehensive income: Assets that are held for collection of contractual cash flows and for selling the financial assets, where the assets' contractual cash flows represent solely payments of principal and interest, are measured at fair value through other comprehensive income. Changes in the carrying amount are recognized in other comprehensive income, except for impairment loss (reversal of impairment loss), interest income, and gains and losses on foreign currency transactions which are recorded in profit or loss. When the financial asset is derecognized, accumulated other comprehensive income is reclassified from equity to profit or loss. Interest income from these financial assets is included in 'finance income' using the effective interest rate method. Gains and losses on foreign currency transactions are presented in 'other income and expenses', and impairment losses are presented in 'other expenses'.
- ✓ Fair value through profit or loss: Assets that do not meet the criteria for amortized cost or fair value through other comprehensive income are measured at fair value through profit or loss. A gain or loss on a debt investment that is subsequently measured at fair value through profit or loss and is not part of a hedging relationship is recognized in profit or loss and presented net in the statement of profit or loss as 'other income and expenses' in the year in which it occurs.

3. Changes in accounting policies, Continued

(3) K-IFRS No. 1109, 'Financial Instruments', continued

② Financial assets, continued

(b) Measurement, continued

(ii) Equity Instruments

The Company subsequently measures all equity investments at fair value. Where the Company's management has elected to present fair value gains and losses on equity investments in other comprehensive income, there is no subsequent reclassification of fair value gains and losses to profit or loss upon derecognition of the investment. Dividend income from such investments continues to be recognized in profit or loss as 'other income' when the right to receive payments is established.

Changes in fair value of financial assets measured at fair value through profit or loss are recognized in 'other income and expenses' in the statement of profit or loss as applicable. Impairment loss (reversal of impairment loss) on equity investments measured at fair value through other comprehensive income is not reported separately from other changes in fair value.

(c) Impairment

The Company assesses the expected credit losses associated with its debt instruments measured at amortized cost or fair value through other comprehensive income based on forward looking information. The impairment methodology applied depends on whether there has been a significant increase in credit risk.

The Company applies the practical expedient when measuring lifetime expected credit losses from initial recognition of trade receivables. There is no impact on the financial statements from estimation of expected credit losses based on actual credit loss experience over the past two years.

② Derivative Instruments

Derivatives are initially recognized at fair value on the date a derivative contract is entered into and are subsequently remeasured at their fair value at the end of each reporting period. The accounting for subsequent changes in fair value depends on whether the derivative is designated as a hedging instrument, and if so, the nature of the item being hedged, and the type of hedge relationship designated.

③ Financial Liabilities

The Company classifies financial liabilities into financial liabilities at fair value through profit or loss or other financial liabilities in accordance with the substance of the contractual arrangement and the definitions of financial liabilities. The Company recognizes financial liabilities in the statements of financial position when the Company becomes a party to the contractual provisions of the financial liability.

(a) Financial liabilities at fair value through profit or loss

Financial liabilities at fair value through profit or loss include financial liabilities held for trading or designated as such upon initial recognition. Subsequent to initial recognition, financial liabilities at fair value through profit or loss are measured at fair value, and changes in fair value are generally expressed as follows:

- Fair value change due to changes in own credit risk : other comprehensive income or loss
- Other fair value change : profit or loss

Upon initial recognition, transaction costs that are directly attributable to the issue are recognized in profit or loss as incurred.

3. Changes in accounting policies, Continued

(3) K-IFRS No. 1109, 'Financial Instruments', continued

③ Financial Liabilities, continued

(b) Other financial liabilities

Non-derivative financial liabilities other than financial liabilities at fair value through profit or loss are classified as other financial liabilities. At the date of initial recognition, other financial liabilities are measured at fair value minus transaction costs that are directly attributable to the issue. Subsequent to initial recognition, other financial liabilities are measured at amortized cost using the effective interest method.

(c) Derecognition of financial liabilities

Financial liabilities are derecognized from the statement of financial position when it is extinguished; for example, when the obligation specified in the contract is discharged, cancelled, or expires.

The Company applied K-IFRS No. 1109 retroactively; however, the financial statements for the year ended December 31, 2016 presented for comparative purpose were not restated. Accordingly, the comparative information is presented in accordance with the previous accounting policy.

4. Significant Accounting Policies

The significant accounting policies applied by the Company in preparation of its financial statements in accordance with K-IFRS are included below. The accounting policies set out below have been applied consistently to all periods presented in these financial statements except for the changes in accounting policies as explained in Note 3.

(1) Interests in equity-accounted investees

The Company's interests in equity-accounted investees comprise interests in associates and joint ventures.

Associates are those entities in which the company has significant influence, but no control or joint control, over the financial and operating policies. A joint venture is an arrangement in which the Company has joint control, whereby the Company has rights to the net assets of the arrangement, rather than rights to its assets and obligations for its liabilities.

Interests in associates and the joint venture are accounted for using the equity method. They are initially recognized at cost, which includes transaction costs. Subsequent to initial recognition, the financial statements include the Company's share of the profit or loss and OCI of equity-accounted investees until the date on which significant influence or joint control ceases.

(2) Cash and cash equivalents

Cash and cash equivalents comprise cash balances and call deposits with maturities of three months or less from the acquisition date that are subject to an insignificant risk of changes in their fair value and are used by the Company in the management of its short-term commitments. Equity investments are excluded from cash equivalents unless they are, in substance, cash equivalents, for example in the case of preferred shares when it has a short maturity with a specified redemption date.

(3) Inventories

The cost of inventories is based on the moving-weighted average principle, except for goods in transit based on specific identification method and includes expenditures for acquiring the inventories, production or conversion costs and other costs incurred in bringing them to their existing location and condition.

4. Significant Accounting Policies, Continued

(3) Inventories, continued

Inventories are measured at the lower of cost and net realizable value. The amount of any write-down of inventories to net realizable value and all losses of inventories are recognized as an expense in the period the write-down or loss occurs. The amount of any reversal of any write-down of inventories, arising from an increase in net realizable value, are recognized as a reduction in the amount of inventories recognized as an expense in the period in which the reversal occurs.

(4) Property, plant and equipment

Property, plant and equipment are initially measured at cost. The cost of property, plant and equipment includes expenditures arising directly from the construction or acquisition of the asset, any costs directly attributable to bringing the asset to the location and condition necessary for it to be capable of operating in the manner intended by management, and the initial estimate of the costs of dismantling and removing the item and restoring the site on which it is located.

Subsequent to initial recognition, an item of property, plant and equipment shall be carried at its cost less any accumulated depreciation and any accumulated impairment losses.

Property, plant and equipment, except for land, are depreciated over estimated useful lives that appropriately reflect the pattern in which the asset's future economic benefits are expected to be consumed on a straight-line basis.

Gains and losses on disposal of an item of property, plant and equipment are determined by comparing the proceeds from disposal with the carrying amount of property, plant and equipment and are recognized in profit or loss.

The estimated useful lives of the property, plant and equipment are as follows:

	Useful lives (years)
Buildings and structures	40
Machinery	5~15
Vehicles	4
Other tangible assets	4

Depreciation methods, useful lives and residual values are reviewed at the end of each reporting date and adjusted, if appropriate. The change is accounted for as a change in an accounting estimate.

(5) Intangible assets

Intangible assets are measured initially at cost and, subsequently, are carried at cost less accumulated amortization and accumulated impairment losses.

Amortization of intangible assets is calculated on a straight-line basis over the following estimated useful lives of intangible assets from the date that they are available for use. The residual value of intangible assets is zero. However, as there are no foreseeable limits to the periods over which memberships are expected to be available for use, they are determined as having indefinite useful lives and not amortized.

	Useful lives (years)
Software	5
Other intangible assets	5
Land usage rights	50
Memberships	indefinite

4. Significant Accounting Policies, Continued

(5) Intangible assets, continued

Useful lives and the amortization methods for intangible assets with finite useful lives are reviewed at the end of each reporting period. The useful lives of intangible assets that are not being amortized are reviewed at the end of each reporting period to determine whether events and circumstances continue to support indefinite useful life assessments for those assets. Changes are accounted for as changes in accounting estimates

Subsequent expenditures are capitalized only when they increase the future economic benefits embodied in the specific asset to which it relates. All other expenditures, including expenditures on internally generated goodwill and brands, are recognized in profit or loss as incurred.

(6) Borrowing costs

The Company capitalizes borrowing costs directly attributable to the acquisition, construction, or production of a qualifying asset as part of the cost of that asset. Other borrowing costs are recognized in expense as incurred. A qualifying asset is an asset that requires a substantial period of time to get ready for its intended use or sale. Financial assets and inventories that are manufactured or otherwise produced over a short period of time are not qualifying assets. Assets that are ready for their intended use or sale when acquired are not qualifying assets.

To the extent that the Company borrows funds specifically for the purpose of obtaining a qualifying asset, the Company determines the amount of borrowing costs eligible for capitalization as the actual borrowing costs incurred on that borrowing during the period less any investment income on the temporary investment of those borrowings. To the extent that the Company borrows funds generally and uses them for the purpose of obtaining a qualifying asset, the Company shall determine the amount of borrowing costs eligible for capitalization by applying a capitalization rate to the expenditures on that asset. The capitalization rate shall be the weighted average of the borrowing costs applicable to the borrowings of the Company that are outstanding during the period, other than borrowings made specifically for the purpose of obtaining a qualifying asset. The amount of borrowing costs that the Company capitalizes during a period shall not exceed the amount of borrowing costs incurred during that period.

(7) Government grants

Government grants are not recognized unless there is reasonable assurance that the Company will comply with the grant's conditions and that the grant will be received.

Government grants whose primary condition is that the Company purchases, constructs or otherwise acquires long-term assets are deducted in calculating the carrying amount of the asset. The grant is recognized in profit or loss over the life of a depreciable asset as a reduction to depreciation expense.

(8) Impairment of non-financial assets

The carrying amounts of the Company's non-financial assets, except assets arising from employee benefits, inventories and deferred tax assets, are reviewed at the end of the reporting period to determine whether there is any indication of impairment. If any such indication exists, then the asset's recoverable amount is estimated. Goodwill and intangible assets that have indefinite useful lives or that are not yet available for use, irrespective of whether there is any indication of impairment, are tested for impairment annually by comparing their recoverable amount to their carrying amount.

For impairment testing, assets are grouped together into the smallest group of assets that generates cash inflows that are largely independent of the cash inflows from other assets or groups of assets ("CGUs"). The recoverable amount of an asset or CGU is the greater of its value in use and its fair value less costs to sell. The value in use is estimated by applying a pre-tax discount rate that reflect current market assessments of the time value of money and the risks specific to the asset or CGU for which estimated future cash flows have not been adjusted, to the estimated future cash flows expected to be generated by the asset or CGU.

An impairment loss is recognized in profit or loss if the carrying amount of an asset or a CGU exceeds its recoverable amount.

4. Significant Accounting Policies, Continued

(9) Leases

The Company classifies and accounts for leases as either a finance or operating lease, depending on the terms. Leases where the Company assumes substantially all of the risks and rewards of ownership are classified as finance leases. All other leases are classified as operating leases.

① Finance leases

At the commencement of the lease term, the Company recognizes as finance assets and finance liabilities in its separate statements of financial position, the lower amount of the fair value of the leased property and the present value of the minimum lease payments, each determined at the inception of the lease. Any initial direct costs are added to the amount recognized as an asset.

② Finance leases, continued

Minimum lease payments are apportioned between the finance charge and the reduction of the outstanding liability. The finance charge is allocated to each period during the lease term so as to produce a constant periodic rate of interest on the remaining balance of the liability. Contingent rents are charged as expenses in the periods in which they are incurred.

The depreciable amount of a leased asset is allocated to each accounting period during the period of expected use on a systematic basis consistent with the depreciation policy the lessee adopts for depreciable assets that are owned. If there is no reasonable certainty that the lessee will obtain ownership by the end of the lease term, the asset is fully depreciated over the shorter of the lease term and its useful life. The Company reviews to determine whether the leased asset may be impaired.

③ Operating leases

Leases where the lessor retains a significant portion of the risks and rewards of ownership are classified as operating leases. Payments made under operating leases (net of any incentives received from the lessor) are recognized in profit or loss on a straight-line basis over the period of the lease.

④ Determining whether an arrangement contains a lease

Determining whether an arrangement is, or contains, a lease shall be based on the substance of the arrangement and requires an assessment of whether fulfillment of the arrangement is dependent on the use of a specific asset or assets (the asset) and the arrangement conveys a right to use the asset.

At inception or reassessment of the arrangement, the Company separates payments and other consideration required by such an arrangement into those for the lease and those for other elements on the basis of their relative fair values. If the Company concludes for a financial lease that it is impracticable to separate the payments reliably, the Company recognizes an asset and a liability at an amount equal to the fair value of the underlying asset that was identified as the subject of the lease. Subsequently, the liability shall be reduced as payments are made and an imputed finance charge on the liability recognized using the purchaser's incremental borrowing rate of interest.

(10) Employee benefits

① Short-term employee benefits

Short-term employee benefits are employee benefits that are expected to be settled wholly before twelve months after the end of the annual reporting period in which the employees render the related services. When an employee has rendered service to the Company during an accounting period, the Company recognizes the undiscounted amount of short-term employee benefits expected to be paid in exchange for that service.

4. Significant Accounting Policies, Continued

(10) Employee benefits, continued

② Other long-term employee benefits

Other long-term employee benefits include employee benefits that are settled beyond 12 months after the end of the period in which the employees render the related service, and are calculated at the present value of the amount of future benefit that employees have earned in return for their service in the current and prior periods. Any changes from re-measurements are recognized through profit or loss in the period in which they arise.

③ Retirement benefits: defined benefit plans

The Company's net obligation in respect of defined benefit plans is calculated separately for each plan by estimating the amount of future benefit that employees have earned in the current and prior periods, discounting that amount and deducting the fair value of any plan assets.

The calculation of defined benefit obligations is performed annually by a qualified actuary using the projected unit credit method. When the calculation results in a potential asset for the Company, the recognized asset is limited to the present value of economic benefits available in the form of any future refunds from the plan or reductions in future contributions to the plan. To calculate the present value of economic benefits, consideration is given to any applicable minimum funding requirements.

Remeasurements of the net defined benefit liability, which comprise actuarial gains and losses, the return on plan assets (excluding interest), and the effect of the asset ceiling (if any, excluding interest), are recognized immediately in OCI. The Company determines the net interest expense (income) on the net defined benefit liability (asset) for the period by applying the discount rate used to measure the defined benefit obligation at the beginning of the annual period to the then-net defined benefit liability (asset), taking into account any changes in the net defined benefit liability (asset) during the period as a result of contributions and benefit payments. Net interest expense and other expenses related to defined benefit plans are recognized in profit or loss.

When the plan amendment or curtailment occurs, gains or losses on amendment or curtailment in benefits for the past service provided are recognized through profit or loss. The Company recognizes gain or loss on a settlement when the settlement of defined benefit plan occurs.

(11) Provisions

Provisions are recognized when the Company has a present legal or constructive obligation as a result of a past event, it is probable that an outflow of resources embodying economic benefits will be required to settle the obligation and a reliable estimate can be made of the amount of the obligation.

The risks and uncertainties that inevitably surround many events and circumstances are taken into account in reaching the best estimate of a provision. Where the effect of the time value of money is material, provisions are determined at the present value of the expected future cash flows.

Where some or all of the expenditures required to settle a provision are expected to be reimbursed by another party, the reimbursement is recognized when, and only when, it is virtually certain that reimbursement will be received if the entity settles the obligation. The reimbursement is presented as a separate asset.

Provisions are reviewed at the end of each reporting period and adjusted to reflect the current best estimates. If it is no longer probable that an outflow of resources embodying economic benefits will be required to settle the obligation, the provision is reversed.

A provision is used only for expenditures for which the provision was originally recognized.

4. Significant Accounting Policies, Continued

(12) Emissions rights

The Company accounts for greenhouse gases emission right and the relevant liability as below pursuant to the *Act on Allocation and Trading of Greenhouse Gas Emission*.

① Greenhouse gases emission right

Greenhouse gases emission right consists of emission allowances which are allocated from the government free of charge or purchased from the market. The cost includes any directly attributable costs incurred during the normal course of business.

Emission rights held for the purpose of performing the obligation is classified as intangible asset and is initially measured at cost. After initial recognition, they are carried at cost less accumulated impairment losses. Emission rights held for short-swing profits are classified as current asset and are measured at fair value with any changes in fair value recognized as profit or loss in the respective reporting period.

The Company derecognizes an emission right asset when the emission allowance is unusable, disposed, or submitted to government in which the future economic benefits are no longer expected to be probable.

② Emission liability

Emission liability is a present obligation of submitting emission rights to the government with regard to emission of greenhouse gas. Emission liability is recognized when it is probable that outflows of resources will be required to settle the obligation and the costs required to perform the obligation are reliably estimable. Emission liability is an amount of estimated obligations for emission rights to be submitted to the government for the performing period. The emission liability is measured based on the expected quantity of emission for the performing period in excess of emission allowance in possession and the unit price for such emission rights in the market at the end of the reporting period.

(13) Foreign currency transactions

Transactions in foreign currencies are translated to the respective functional currency of the Company at exchange rates at the dates of the transactions. Monetary assets and liabilities denominated in foreign currencies are retranslated to the functional currency using the reporting date's exchange rate. Non-monetary assets and liabilities denominated in foreign currencies that are measured at fair value are retranslated to the functional currency at the exchange rate at the date that the fair value was determined.

Foreign currency differences arising on retranslation are recognized in profit or loss, except for differences arising on the retranslation of available-for-sale equity instruments, a financial liability designated as a hedge of the net investment in a foreign operation, or qualifying cash flow hedges, which are recognized in other comprehensive income.

(14) Equity capital

Ordinary shares are classified as equity. Incremental costs directly attributable to the issuance of ordinary shares and share options are recognized as a deduction from equity, net of any tax effects.

(15) Revenue

Revenue from the sale of goods, rendering of services or use of the Company assets is measured at the fair value of the consideration received or receivable, net of returns, trade discounts and volume rebates. Revenue is recognized when persuasive evidence exists, usually in the form of an executed sales agreement, that the significant risks and rewards of ownership have been transferred to the buyer, recovery of the consideration is probable, the associated costs and possible return of goods can be estimated reliably, there is no continuing management involvement with the goods, and the amount of revenue can be measured reliably. The timing of the transfers of risks and rewards varies depending on the individual terms of the contract of sale.

4. Significant Accounting Policies, Continued

(15) Revenue, continued

When two or more revenue generating activities or deliverables are sold under a single arrangement, each deliverable that is considered to be a separate unit of account is accounted for separately. The allocation of consideration from a revenue arrangement to its separate units of account is based on the relative fair values of each unit. Revenue from services rendered is recognized in profit or loss in proportion to the stage of completion of the transaction at the reporting date. The stage of completion is assessed by reference to surveys of work performed.

(16) Finance income and finance costs

Finance income comprises interest income on funds invested. Interest income is recognized as it accrues in profit or loss, using the effective interest method.

Finance costs comprise interest expense on borrowings and changes in the fair value of derivatives financial instruments at fair value through profit or loss. Borrowing costs that are not directly attributable to the acquisition, construction, or production of a qualifying asset are recognized in profit or loss using the effective interest method.

(17) Income taxes

Income tax expense comprises current and deferred tax. Current tax and deferred tax are recognized in profit or loss except to the extent that it relates to a business combination or items recognized directly in equity or in other comprehensive income.

If the Company determines that interests and penalties from income taxes fall under income taxes, it accounts them for in accordance with K-IFRS No. 1012 'Income taxes'. If not, then it applies K-IFRS 1037 'Provisions, Contingent Liabilities and Contingent Assets'

① Current tax

Current tax is the expected tax payable or receivable on the taxable profit or loss for the year, using tax rates enacted or substantively enacted at the end of the reporting period and any adjustment to tax payable in respect of previous years. The taxable profit is different from the accounting profit for the period since the taxable profit is calculated excluding the temporary differences, which will be taxable or deductible in determining taxable profit (tax loss) of future periods, and non-taxable or non-deductible items from the accounting profit.

② Deferred tax

Deferred tax is recognized, using the asset-liability method, in respect of temporary differences between the carrying amounts of assets and liabilities for financial reporting purposes and the amounts used for taxation purposes. A deferred tax liability is recognized for all taxable temporary differences. A deferred tax asset is recognized for all deductible temporary differences to the extent that it is probable that taxable profit will be available against which they can be utilized. However, deferred tax is not recognized for taxable temporary differences arising on the initial recognition of goodwill, or the initial recognition of assets or liabilities in a transaction that is not a business combination and that affects neither accounting profit or loss nor taxable income.

If future taxable profit is likely to occur, unused tax losses, tax credits, and deductible temporary differences are recognized as deferred tax assets within a range of future taxable profit. Future taxable profit is determined from the expiration of related taxable temporary differences. If taxable temporary difference is not sufficient to fully recognize the deferred tax asset, the expiration of temporary differences will be considered.

The measurement of deferred tax liabilities and deferred tax assets reflects the tax consequences that would follow from the manner in which the Company expects, at the end of the reporting period, to recover or settle the carrying amount of its assets and liabilities.

4. Significant Accounting Policies, Continued

(17) Income taxes, continued

② Deferred tax, continued

The Company recognizes a deferred tax liability for all taxable temporary differences associated with investments in subsidiaries, associates, and interests in joint ventures, except to the extent that the Company is able to control the timing of the reversal of the temporary difference and it is probable that the temporary difference will not reverse in the foreseeable future. The Company recognizes a deferred tax asset for all deductible temporary differences arising from investments in subsidiaries and associates, to the extent that it is probable that the temporary difference will reverse in the foreseeable future and taxable profit will be available against which the temporary difference can be utilized.

The carrying amount of a deferred tax asset is reviewed at the end of each reporting period and reduced to the extent that it is no longer probable that sufficient taxable profit will be available to allow the benefit of part or all of that deferred tax asset to be utilized.

Deferred tax assets and liabilities are measured at the tax rates that are expected to apply to the period when the asset is realized or the liability is settled, based on tax rates (and tax laws) that have been enacted or substantively enacted by the end of the reporting period.

Deferred tax assets and liabilities are offset only if there is a legally enforceable right to the Company to offset the related current tax liabilities and assets, they relate to income taxes levied by the same tax authority, and the Company intends to settle current tax liabilities and assets on a net basis.

(18) Earnings per share

The Company presents basic and diluted earnings per share (EPS) data for its ordinary shares. Basic EPS is calculated by dividing the profit or loss attributable to ordinary shareholders of the Company by the weighted average number of ordinary shares outstanding during the period, adjusted for own shares held. The Company has no dilutive potential shares, therefore diluted loss per share is equal to the basic earnings (loss) per share.

(19) Significant accounting policies for financial statements for prior year presented for comparative figures

① Non-derivative financial assets

The Company recognizes and measures non-derivative financial assets by the following four categories: financial assets at fair value through profit or loss, held-to-maturity investments, loans and receivables, and available-for-sale financial assets. The Company recognizes financial assets in the separate statement of financial position when the Company becomes a party to the contractual provisions of the instrument.

Upon initial recognition, non-derivative financial assets are measured at their fair value plus, in the case of a financial asset not at fair value through profit or loss, transaction costs that are directly attributable to the asset's acquisition or issuance.

(a) Financial assets at fair value through profit or loss

A financial asset is classified as financial assets at fair value through profit or loss if it is held for trading or is designated as such upon initial recognition. Upon initial recognition, transaction costs are recognized in profit or loss when incurred. Financial assets at fair value through profit or loss are measured at fair value, and changes therein are recognized in profit or loss.

(b) Held-to-maturity investments

A non-derivative financial asset with a fixed or determinable payment and fixed maturity, for which the Company has the positive intention and ability to hold to maturity, is classified as held-to-maturity investments. Subsequent to initial recognition, held-to-maturity investments are measured at amortized cost using the effective interest method.

4. Significant Accounting Policies, Continued

(19) Significant accounting policies for financial statements for prior year presented for comparative figures, continued

① Non-derivative financial assets, continued

(c) Loans and receivables

Loans and receivables are financial assets with fixed or determinable payments that are not quoted in an active market. Subsequent to initial recognition, loans and receivables are measured at amortized cost using the effective interest method except for loans and receivables of which the effect of discounting is immaterial.

(d) Available-for-sale financial assets

Available-for-sale financial assets are those non-derivative financial assets that are designated as available-for-sale or are not classified as financial assets at fair value through profit or loss, held-to-maturity investments, or loans and receivables. Subsequent to initial recognition, they are measured at fair value, which changes in fair value, net of any tax effect, recorded in accumulated other comprehensive income in equity. Investments in equity instruments that do not have a quoted market price in an active market and whose fair value cannot be reliably measured and derivatives that are linked to and must be settled by delivery of such unquoted equity instruments are measured at cost.

(e) De-recognition of financial assets

The Company derecognizes a financial asset when the contractual rights to the cash flows from the asset expire, or it transfers the rights to receive the contractual cash flows on the financial asset in a transaction in which substantially all the risks and rewards of ownership of the financial asset are transferred. Any interest in transferred financial assets that is created or retained by the Company is recognized as a separate asset or liability.

If the Company retains substantially all the risks and rewards of ownership of the transferred financial assets, the Company continues to recognize the transferred financial assets and recognizes financial liabilities for the consideration received.

(f) Offsetting between financial assets and financial liabilities

Financial assets and financial liabilities are offset and the net amount is presented in the separate statement of financial position only when the Company currently has a legally enforceable right to offset the recognized amounts, and there is the intention to settle on a net basis or to realize the asset and settle the liability simultaneously.

② Derivative financial instruments

Derivatives are initially recognized at fair value except an effective portion of a derivative hedging instrument. Subsequent to initial recognition, derivatives are measured at fair value, and changes therein are accounted for profit or loss.

③ Impairment of financial assets

A financial asset not carried at fair value through profit or loss is assessed at each reporting date to determine whether there is objective evidence that it is impaired. A financial asset is impaired if objective evidence indicates that a loss event has occurred after the initial recognition of the asset, and that the loss event had a negative effect on the estimated future cash flows of that asset that can be estimated reliably. However, losses expected as a result of future events, regardless of likelihood, are not recognized.

If there is objective evidence that financial instruments are impaired, impairment losses are measured and recognized. In addition, for an investment in an equity security, a significant or prolonged decline in its fair value below its cost is objective evidence of impairment.

4. Significant Accounting Policies, Continued

(19) Significant accounting policies for financial statements for prior year presented for comparative figures, continued

③ Impairment of financial assets

(a) Financial assets measured at amortized cost

An impairment loss in respect of a financial asset measured at amortized cost is calculated as the difference between its carrying amount and the present value of its estimated future cash flows discounted at the asset's original effective interest rate. Losses are recognized in profit or loss and reflected in an allowance account. When the Company considers that there are no realistic prospects of recovery of the asset, the relevant amounts are written off. If the amount of impairment loss subsequently decreases and the decrease can be related objectively to an event occurring after the impairment was recognized, then the previously recognized impairment loss is reversed through profit or loss either directly or by adjusting an allowance account.

(b) Financial assets carried at cost

If there is objective evidence that an impairment loss has occurred on an unquoted equity instrument that is not carried at fair value because its fair value cannot be reliably measured, or on a derivative asset that is linked to and must be settled by delivery of such an unquoted equity instrument, the amount of the impairment loss is measured as the difference between the carrying amount of the financial asset and the present value of estimated future cash flows discounted at the current market rate of return for a similar financial asset. Such impairment losses shall not be reversed.

(c) Available-for-sale financial assets

When a decline in the fair value of an available-for-sale financial asset has been recognized in other comprehensive income and there is objective evidence that the asset is impaired, the cumulative loss that had been recognized in other comprehensive income shall be reclassified from equity to profit or loss as a reclassification adjustment even though the financial asset has not been derecognized. Impairment losses recognized in profit or loss for an investment in an equity instrument classified as available-for-sale shall not be reversed through profit or loss. If, in a subsequent period, the fair value of a debt instrument classified as available-for-sale increases and the increase can be objectively related to an event occurring after the impairment loss was recognized in profit or loss, the impairment loss shall be reversed with the amount of the reversal recognized in profit or loss.

④ Non-derivative financial liabilities

The Company classifies non-derivative financial liabilities into financial liabilities at fair value through profit or loss or other financial liabilities in accordance with the substance of the contractual arrangement and the definitions of financial liabilities. The Company recognizes financial liabilities in the statement of financial position when the Company becomes a party to the contractual provisions of the financial liability.

(a) Financial liabilities at fair value through profit or loss

Financial liabilities at fair value through profit or loss include financial liabilities held for trading or designated as such upon initial recognition. Subsequent to initial recognition, financial liabilities at fair value through profit or loss are measured at fair value, and changes therein are recognized in profit or loss. Upon initial recognition, transaction costs that are directly attributable to the acquisition are recognized in profit or loss as incurred.

(b) Other financial liabilities

Non-derivative financial liabilities other than financial liabilities at fair value through profit or loss are classified as other financial liabilities. At the date of initial recognition, other financial liabilities are measured at fair value minus transaction costs that are directly attributable to the acquisition. Subsequent to initial recognition, other financial liabilities are measured at amortized cost using the effective interest method.

(c) Derecognition of financial liabilities

The Company derecognizes a financial liability from the statement of financial position when it is extinguished (i.e. when the obligation specified in the contract is discharged, cancelled or expires).

4. Significant Accounting Policies, Continued

(20) New standards and interpretations not yet adopted

The following new standards, interpretations and amendments to existing standards have been published and are mandatory for the Company for annual periods beginning on January 1, 2018, and the Company has not early adopted them.

① K-IFRS No. 1115 'Revenue from Contracts with Customers'

K-IFRS No. 1115 'Revenue from Contracts with Customers', published on November 6, 2015, is effective for annual periods beginning on or after January 1, 2018, with earlier adoption permitted. It replaces existing revenue recognition guidance, including K-IFRS No. 1018 'Revenue', K-IFRS No. 1011 'Construction Contracts', K-IFRS No. 2031 'Revenue- Barter transactions involving advertising services', K-IFRS No. 2113 'Customer Loyalty Programs', K-IFRS No. 2115 'Agreements for the construction of real estate', and K-IFRS No. 2118 'Transfers of assets from customers'.

The Company assessed the adoption effect of K-IFRS No. 1115 through analysis from external experts. As of December 31, 2017, the Company is in the inquiring process with Korea Accounting Institute for accounting issues, such as identification of performance obligations on the certification services of manufacturing facilities and timing of revenue recognition. The Company will confirm accounting policies through discussions with related institutions and analyze the detailed financial impact of adopting K-IFRS No. 1115. Expected impacts on the financial statements are generally categorized as follows:

(a) Sale of goods

Revenue is recognized when persuasive evidence exists, usually in the form of an executed sales agreement, that the significant risks and rewards of ownership have been transferred to the buyer, collection of the consideration is probable, the associated costs and possible return of goods can be estimated reliably, there is no continuing involvement with the goods, and the amount of revenue can be measured reliably.

In accordance with K-IFRS No. 1115, the Company recognizes revenue when a customer obtains control of the goods. In the case of biopharmaceuticals, control is transferred when the production of the product is completed, and then it obtains customer's quality approval(Q/R, Quality Release). The Company recognizes revenue at a point of customer's Q/R.

(b) Services rendered

The Company provides storage service for the goods. If a single contract includes a number of performance obligations, or if each performance obligation is performed over a different reporting period, the transaction price is allocated to the performance obligation based on the relative fair value. Revenue from storage service rendered is recognized in proportion to the stage of completion of the transaction at the reporting date.

In accordance with K-IFRS No. 1115, the total consideration of a contract is allocated based on the individual selling prices. The individual selling price will be determined by the stated price at which the Company sells the service in an individual transaction.

The Company believes that fair value and the individual selling price of the service will be similar. Therefore, the Company does not expect that the adoption of K-IFRS No. 1115 will result in a significant difference in timing of revenue recognition for these services.

3) Transition

The Company intends to apply the retrospective approach which will recognize the cumulative impact of initially applying the revenue standard as of January 1, 2018, the date of initial application. Accordingly, upon adoption of K-IFRS No. 1115, the Company will not restate the financial statements for comparative periods.

4. Significant Accounting Policies, Continued

(20) New standards and interpretations not yet adopted, continued

② K-IFRS No.2122, 'Foreign Currency Transactions and Advance Consideration'

According to the enacted Interpretation, the date of transaction for the purpose of determining the exchange rate to use on initial recognition of the related asset, expense or income (or part of it) is the date on which an entity initially recognizes the non-monetary asset or non-monetary liability arising from the prepayment or receipt of advance consideration. In case there are multiple payments or receipts in advance, the entity should determine a date of the transaction for each payment or receipt of advance consideration. This enactment is effective for annual periods beginning on or after January 1, 2018 with early adoption permitted. The Company plans to adopt this enactment for the year beginning on January 1, 2018 and does not expect the enactment to have a significant impact on the financial statements.

③ K-IFRS No.1116, 'Leases'

K-IFRS No. 1116 'Leases' will replace K-IFRS No. 1017 'Leases' and K-IFRS No. 2104 'Determining whether an Arrangement contains a Lease'.

It is effective for annual periods beginning on or after January 1, 2019, with early adoption permitted for a company which has adopted K-IFRS No. 1115.

K-IFRS No. 1116 suggests a single accounting model that requires a lessee to recognize lease related asset and liability in the financial statements. A lessee is required to recognize a right-of-use asset representing its right to use the underlying leased asset and a lease liability representing its obligation to make lease payments. The lessee may elect not to apply the requirements to short-term lease of which has a term of 12 months or less at the commencement date and low value assets. Accounting treatment for lessor is similar to the existing standard which classifies lease into finance and operating lease.

The Company started initial assessment on the adoption of K-IFRS No.1116; however, more detailed assessment has not been conducted. The actual financial impact at the date of initial adoption is determined by a future economic environments at the date of initial application such as the borrowing interest rate and a portfolio of lease contracts as of January 1, 2019, execution of lease renewal option and coverage in use of a practical expedient and lease recognition exemption and others.

As a result of the adoption of K-IFRS No.1116, the nature of expense recognized in relation with leases will be changed as the operating lease expenses previously recognized on a straight-line basis will be changed to depreciation expenses of right-of-use assets and interest expenses of lease liabilities.

In accordance with K-IFRS No.1116, new assets and liabilities should be recognized for operating lease assets except for short-term lease with a contract term of 12-month or less at the commencement date and assets with low value. However, the Company expects that there will be no significant impact on the financial statements because most of leases the Company contracted are short-term lease with a lease-term of 12-month or less and for assets at low value.

As a lessee, the Company can choose to apply the K-IFRS No.1116 using either:

- a full retrospective approach; or
- a modified retrospective approach with a practical expedient.

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2017 and 2016

4. Significant Accounting Policies, Continued

(20) New standards and interpretations not yet adopted, continued

③ K-IFRS No.1116, 'Leases', continued

A lessee should apply one of the approach consistently for entire lease contracts. The Company plans to use a modified retrospective approach, which requires cumulative impacts to be recognized in retained earnings as of January 1, 2019 and does not restate the comparative financial statements for comparative periods.

If the Company is to use a modified retrospective approach to leases classified as operating lease under K-IFRS No.1017, it, as a lessee, can select by lease contracts whether to apply a practical expedient. The Company plans to assess potential impact of applying such practical expedient.

When modified retrospective approach is applied for the lease contracts classified as an operating lease under K-IFRS No.1017, lessee may elect the application of various practical expedients for each existing lease contracts at the date of adoption of K-IFRS No.1116. The Company will assess the financial impact when the practical expedient is applied.

5. Operating Segments

(1) The Company has disclosed operating segments in the financial statements in accordance with K-IFRS No. 1108. The Company has one reportable segment, 'Biosimilar manufacturing', as of December 31, 2017.

(2) Amount of revenue for three customers who take over 10% or more of the Company's total revenue is ₩440,951 million for the year ended December 31, 2017 (₩292,833 million for three customers for the year ended December 31, 2016).

(3) Geographic revenue for the years ended December 31, 2017 and 2016 were as follows:

(In thousands of won)

	2017	2016
Domestic	₩ 18,610,857	929,634
Europe	440,951,087	292,832,778
Other	5,067,390	859,609
	₩ 464,629,334	294,622,021

6. Financial Instruments

(1) Financial assets

Categories of financial assets as of December 31, 2017 and 2016 are as follows:

(In thousands of won)

	December 31, 2017	December 31, 2016
	Financial assets measured at amortized cost	Loans and receivables
Current assets		
Cash and cash equivalents	₩ 358,088,091	230,337,601
Short-term financial instruments	-	970,000,000
Trade and other receivables	25,817,150	75,630,820
Deposits provided	22,605,000	16,289,300
	406,510,241	1,292,257,721
Non-current assets		
Long-term financial instruments	7,000	8,000
Deposits provided	20,560,000	21,475,000
	20,567,000	21,483,000
₩	427,077,241	1,313,740,721

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2017 and 2016

6. Financial Instruments, Continued

(2) Financial liabilities

Categories of financial liabilities as of December 31, 2017 and 2016 are as follows:

(In thousands of won)

	December 31, 2017			December 31, 2016		
	Financial liabilities at fair value through profit or loss	Financial liabilities measured at amortized cost	Total	Financial liabilities through profit or loss	Financial liabilities measured at amortized cost	Total
Current liabilities						
Trade and other payables	₩ -	158,214,846	158,214,846	-	138,056,375	138,056,375
Derivative liabilities(*)	1,933,562,427	-	1,933,562,427	1,874,470,943	-	1,874,470,943
Bonds	-	49,926,578	49,926,578	-	169,917,877	169,917,877
Short-term borrowings	-	40,000,000	40,000,000	-	-	-
Current portion of long-term borrowings	-	48,750,000	48,750,000	-	233,776,207	233,776,207
	<u>1,933,562,427</u>	<u>296,891,424</u>	<u>2,230,453,851</u>	<u>1,874,470,943</u>	<u>541,750,459</u>	<u>2,416,221,402</u>
Non-current liabilities						
Trade and other payables	-	2,650,638	2,650,638	-	2,300,380	2,300,380
Bonds	-	50,000,000	50,000,000	-	50,000,000	50,000,000
Long-term borrowings	-	357,235,570	357,235,570	-	389,363,194	389,363,194
	<u>-</u>	<u>409,886,208</u>	<u>409,886,208</u>	<u>-</u>	<u>441,663,574</u>	<u>441,663,574</u>
	<u>₩ 1,933,562,427</u>	<u>706,777,632</u>	<u>2,640,340,059</u>	<u>1,874,470,943</u>	<u>983,414,033</u>	<u>2,857,884,976</u>

(*) The Company recognized the fair value of the call option on Samsung Bioepis Co., Ltd.'s shares held by Biogen Therapeutics Inc. as derivative liabilities. The financial derivative liabilities will expire in 2018, and the option price is the sum of the principal amount and accumulated interest. The value of the financial derivative liabilities may be changed significantly depending on the market situation and the exercise of the option.

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2017 and 2016

6. Financial Instruments, Continued

(3) Reclassification of financial instruments due to adoption of K-IFRS No.1109

On the date of initial application of K-IFRS No.1109, January 1, 2017, the Company has classified its financial instruments into the appropriate K-IFRS No.1109 categories. The main effects resulting from this reclassification are as follows:

(In thousands of won)

	Categories		Carrying amount		
	K-IFRS No.1039	K-IFRS No.1109	K-IFRS No.1039	K-IFRS No.1109	Difference
Current financial asset:					
Cash and cash equivalents	Loans and receivables	Financial assets measured at amortized cost	₩ 230,337,601	230,337,601	-
Short-term financial instruments			970,000,000	970,000,000	-
Trade and other receivables			75,630,820	75,630,820	-
Deposits provided			16,289,300	16,289,300	-
Non-current financial asset:					
Long-term financial instruments	Loans and receivables	Financial assets measured at amortized cost	8,000	8,000	-
Deposits provided			21,475,000	21,475,000	-
Current financial liabilities:					
Derivative liabilities	Financial liabilities through profit or loss	Financial liabilities at fair value through profit or loss	1,874,470,943	1,874,470,943	-
Trade and other payables	Financial liabilities measured at amortized cost	Financial liabilities measured at amortized cost	138,056,375	138,056,375	-
Bonds			169,917,877	169,917,877	-
Current portion of long-term borrowings			233,776,207	233,776,207	-
Non-current financial liabilities:					
Trade and other payables	Financial liabilities measured at amortized cost	Financial liabilities measured at amortized cost	2,300,380	2,300,380	-
Bonds			50,000,000	50,000,000	-
Borrowings			389,363,194	389,363,194	-

(4) Fair value hierarchy

(i) The fair value hierarchy of financial instruments carried at fair value as of December 31, 2017 is as follows:

(In thousands of won)

	Level 1	Level 2	Level 3	Total
Derivative liabilities	₩ -	-	1,933,562,427	1,933,562,427

(ii) The fair value hierarchy of financial instruments carried at fair value as of December 31, 2016 is as follows:

(In thousands of won)

	Level 1	Level 2	Level 3	Total
Derivative liabilities	₩ -	-	1,874,470,943	1,874,470,943

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2017 and 2016

6. Financial Instruments, Continued

(5) Valuation techniques and inputs

(i) Valuation techniques and significant inputs used in measuring the fair value of financial instruments classified as level 3 as of December 31, 2017 are as follows:

(In thousands of won)

		Fair Value	Valuation techniques	Significant unobservable inputs
Derivative liabilities	₩	1,933,562,427	Binomial model	Fair value of underlying asset (Samsung Bioepis Co., Ltd.)

(ii) Valuation techniques and significant inputs used in measuring the fair value of financial instruments classified as level 3 as of December 31, 2016 are as follows:

(In thousands of won)

		Fair Value	Valuation techniques	Significant unobservable inputs
Derivative liabilities	₩	1,874,470,943	Binomial model	Fair value of underlying asset (Samsung Bioepis Co., Ltd.)

For the fair values of derivative liabilities, reasonably possible changes at the reporting date to one of the significant unobservable inputs, holding other inputs constant, would have the following effects.

(In thousands of won)

Inputs(*)	Volatility		Favorable changes	Unfavorable changes
Weighted average cost of capital	0.5%	₩	172,739,671	(192,298,935)
Permanent growth rate	0.5%		98,386,789	(108,747,664)

(*)Changes in fair value of derivative liabilities is calculated by increasing or decreasing significant unobservable inputs used in fair value measurement of the underlying assets.

(6) Finance income and costs by categories

Details of finance income and costs for the years ended December 31, 2017 and 2016 are as follows:

(In thousands of won)

		Interest income and costs		Loss on valuation of derivatives		Other income and costs (*)		Net income (costs)	
		2017	2016	2017	2016	2017	2016	2017	2016
Financial assets measured at amortized cost	₩	9,949,872	3,639,650	-	-	(9,359,709)	1,173,708	590,163	4,813,358
Financial liabilities measured at amortized cost		(5,683,983)	(19,254,372)	-	-	11,059,747	(3,961,606)	5,375,764	(23,215,978)
Financial liabilities at fair value through profit or loss		-	-	(59,091,484)	(54,031,894)	-	-	(59,091,484)	(54,031,894)
	₩	<u>4,265,889</u>	<u>(15,614,722)</u>	<u>(59,091,484)</u>	<u>(54,031,894)</u>	<u>1,700,038</u>	<u>(2,787,898)</u>	<u>(53,125,557)</u>	<u>(72,434,514)</u>

(*) Other income and costs include gain (loss) on foreign currency translations and transactions.

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2017 and 2016

7. Cash and Cash Equivalents

Cash and cash equivalents as of December 31, 2017 and 2016 are as follows:

(In thousands of won)

		December 31, 2017	December 31, 2016
Demand deposits	₩	358,088,091	30,337,601
Other cash equivalents		-	200,000,000
	₩	<u>358,088,091</u>	<u>230,337,601</u>

8. Restricted Financial Instruments

Restriction of financial instruments as of December 31, 2017 and 2016 are as follows:

(In thousands of won)

		December 31, 2017	December 31, 2016	Note
Long-term financial instruments	₩	7,000	8,000	Deposits on checking accounts

9. Trade and Other Receivables

(1) Details of trade and other receivables as of December 31, 2017 and 2016 are as follows:

(In thousands of won)

		December 31, 2017			December 31, 2016		
		Related parties	Others	Total	Related parties	Others	Total
Trade receivables	₩	2,503,684	16,015,102	18,518,786	-	66,461,789	66,461,789
Other receivables		1,165,506	5,296,165	6,461,671	100,208	5,922,049	6,022,257
Accrued income		579,605	257,088	836,693	19,678	3,127,096	3,146,774
	₩	<u>4,248,795</u>	<u>21,568,355</u>	<u>25,817,150</u>	<u>119,886</u>	<u>75,510,934</u>	<u>75,630,820</u>

(2) The aging analyses of trade and other receivables that past due but not impaired as of December 31, 2017 and 2016 are as follows:

(In thousands of won)

		December 31, 2017	December 31, 2016
Less than 1 month (*)	₩	24,762,078	18,339,121
More than 1 month, less than 6 months (*)		1,054,998	57,142,373
More than 6 months, less than 12 months (*)		72	138,608
More than 12 months (*)		2	10,718
	₩	<u>25,817,150</u>	<u>75,630,820</u>

(*) It means that the period has passed from the day of receivables issued.

The Company calculated an estimate of expected credit losses based on actual credit loss experience over the past two years and determined that it is not necessary to recognize allowance for trade and other receivables.

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2017 and 2016

10. Inventories

(1) Details of inventories as of December 31, 2017 are as follows:

(In thousands of won)

		December 31, 2017		
		Acquisition cost	Provision for loss on valuation	Carrying amounts
Finished goods	₩	104,077,102	(8,793,628)	95,283,474
Work-in-process		22,128,639	(6,770)	22,121,869
Raw materials		45,196,129	-	45,196,129
Supplies		47,385,245	-	47,385,245
	₩	<u>218,787,115</u>	<u>(8,800,398)</u>	<u>209,986,717</u>

(2) Details of inventories as of December 31, 2016 are as follows:

(In thousands of won)

		December 31, 2016		
		Acquisition cost	Provision for loss on valuation	Carrying amounts
Finished goods	₩	91,542,244	(8,413,216)	83,129,028
Work-in-process		13,610,021	(2,425,743)	11,184,278
Raw materials		35,630,578	-	35,630,578
Supplies		34,276,096	-	34,276,096
	₩	<u>175,058,939</u>	<u>(10,838,959)</u>	<u>164,219,980</u>

(3) During the years ended December 31, 2017 and 2016, provision for loss on valuation of ₩2,038,561 thousand and ₩1,275,392 thousand was reversed respectively. The amounts are included in cost of revenue.

11. Other Assets

Details of other assets as of December 31, 2017 and 2016 are as follows:

(In thousands of won)

		December 31, 2017		December 31, 2016	
		Current	Non-current	Current	Non-current
Prepaid income taxes	₩	17,076	-	149,511	-
Advance payments		140,251	-	23,189	-
Prepaid expenses		4,340,850	963,402	4,774,776	747,220
	₩	<u>4,498,177</u>	<u>963,402</u>	<u>4,947,476</u>	<u>747,220</u>

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2017 and 2016

12. Investments in Associates and Joint Ventures

(1) Investments in associates and joint ventures as of December 31, 2017 and 2016 are as follows:

(In thousands of won)

Investee	Major Business location	Primary business	Fiscal date	December 31, 2017		December 31, 2016	
				Carrying amount	Percentage of ownership	Carrying amount	Percentage of ownership
Archigen Biotech Limited	United Kingdom	Bio-medicine business	December 31, 2017 ₩	10,060,921	50.00%	48,832,293	50.00%
Samsung Bioepis Co., Ltd.(*)	Republic of Korea	Biosimilar research and commercialization	December 31, 2017	4,998,173,084	94.61%	4,895,523,997	93.31%
				₩ 5,008,234,005		4,944,356,290	

(*) Samsung Bioepis Co., Ltd. was set up with Biogen Therapeutics Inc. in 2012. Samsung Bioepis Co., Ltd. had been classified as a subsidiary until 2014. During 2015, as potential voting rights held by Biogen Therapeutics Inc. were changed to substantive rights to Samsung Bioepis Co., Ltd., the Company lost control over Samsung Bioepis Co., Ltd. If Biogen Therapeutics Inc. does not exercise the option until the expiration (in 2018), the judgment of control could be changed.

(2) Changes in investments in associates and joint ventures for the year ended December 31, 2017 and 2016 are as follows:

(In thousands of won)

Investee	2017				
	Beginning balance	Acquisition	Equity loss	Equity change	Ending balance
Archigen Biotech Limited ₩	48,832,293	-	(31,664,555)	(7,106,817)	10,060,921
Samsung Bioepis Co., Ltd.	4,895,523,997	200,014,156	(97,937,783)	572,714	4,998,173,084
₩	4,944,356,290	200,014,156	(129,602,338)	(6,534,103)	5,008,234,005

(In thousands of won)

Investee	2016				
	Beginning balance	Acquisition	Equity loss	Equity change	Ending balance
Archigen Biotech Limited ₩	28,961,024	34,653,000	(17,477,034)	2,695,303	48,832,293
Samsung Bioepis Co., Ltd.	4,808,578,367	200,000,000	(111,745,616)	(1,308,754)	4,895,523,997
₩	4,837,539,391	234,653,000	(129,222,650)	1,386,549	4,944,356,290

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2017 and 2016

12. Investments in Associates and Joint Ventures, Continued

(3) Financial information of equity-accounted investees as of and for the years ended December 31, 2017 and 2016 is summarized as follows:

(In thousands of won)

December 31, 2017							
Investee	Current assets	Non-current assets	Current liabilities	Non-current liabilities	Revenue	Net loss	Total comprehensive loss
Archigen Biotech Limited ₩	24,153,045	75,754,938	19,664,719	1,884,975	-	(63,190,281)	(63,190,281)
Samsung Bioepis Co., Ltd.(*)	547,933,926	637,765,738	453,562,018	237,239,161	314,756,789	(88,346,279)	(87,644,938)

(*) The amounts are based on consolidated financial information of Samsung Bioepis Co., Ltd. before adjustment of fair value of net assets.

(In thousands of won)

December 31, 2016							
Investee	Current assets	Non-current assets	Current liabilities	Non-current liabilities	Revenue	Net loss	Total comprehensive loss
Archigen Biotech Limited ₩	73,238,975	90,513,997	7,690,968	160,970	-	(34,939,862)	(34,939,862)
Samsung Bioepis Co., Ltd.(*)	474,655,563	570,584,078	573,332,338	89,267,849	147,476,750	(106,459,587)	(107,796,326)

(*) The amounts are based on consolidated financial information of Samsung Bioepis Co., Ltd. before adjustment of fair value of net assets.

(4) Reconciliation of the investees' financial information to the carrying amount of the Company's interest in associates and joint ventures as of December 31, 2017 and 2016 is as follows:

(In thousands of won)

December 31, 2017						
Investee	Net assets	Percentage of ownership	Company's share of net assets	Investment difference(*)	Elimination of unrealized profit	Carrying amount
Archigen Biotech Limited ₩	78,358,290	50.00%	39,179,145	-	(29,118,224)	10,060,921
Samsung Bioepis Co., Ltd.	495,357,294	94.61%	468,635,208	4,529,537,876	-	4,998,173,084

(*) Investment difference includes adjustment of net assets value at point of loss of control and an amount related to irregular capital increase.

(In thousands of won)

December 31, 2016						
Investee	Net assets	Percentage of ownership	Company's share of net assets	Investment difference(*)	Elimination of unrealized profit	Carrying amount
Archigen Biotech Limited ₩	155,901,033	50.00%	77,950,517	-	(29,118,224)	48,832,293
Samsung Bioepis Co., Ltd.	382,639,454	93.31%	357,049,036	4,538,474,961	-	4,895,523,997

(*) Investment difference includes adjustment of net assets value at point of loss of control and a tentative amount related to irregular capital increase.

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2017 and 2016

13. Property, Plant and Equipment

(1) Details of property, plant and equipment as of December 31, 2017 and 2016 are as follows:

(In thousands of won)

		December 31, 2017		
		Acquisition cost	Accumulated depreciation	Carrying amount
Buildings	₩	615,351,047	(25,952,623)	589,398,424
Structures		7,731,647	(467,049)	7,264,598
Machinery		729,333,461	(132,970,776)	596,362,685
Vehicles		915,080	(428,264)	486,816
Other tangible assets		22,265,352	(11,289,819)	10,975,533
Construction-in-progress		308,844,562	-	308,844,562
	₩	<u>1,684,441,149</u>	<u>(171,108,531)</u>	<u>1,513,332,618</u>

(In thousands of won)

		December 31, 2016		
		Acquisition cost	Accumulated depreciation	Carrying amount
Buildings	₩	294,885,656	(16,907,193)	277,978,463
Structures		7,479,077	(279,020)	7,200,057
Machinery		643,893,848	(82,443,594)	561,450,254
Vehicles		451,695	(346,133)	105,562
Other tangible assets		18,817,867	(7,297,897)	11,519,970
Construction-in-progress		232,705,460	-	232,705,460
	₩	<u>1,198,233,603</u>	<u>(107,273,837)</u>	<u>1,090,959,766</u>

(2) Changes in property, plant and equipment for the years ended December 31, 2017 and 2016, are as follows:

(In thousands of won)

		2017					
		Beginning balance	Acquisition	Disposals	Depreciation	Others	Ending balance
Buildings	₩	277,978,463	3,516,171	-	(9,069,308)	316,973,098	589,398,424
Structures		7,200,057	-	-	(188,029)	252,570	7,264,598
Machinery		561,450,254	446,636	(2,727)	(50,527,847)	84,996,369	596,362,685
Vehicles		105,562	463,385	-	(82,131)	-	486,816
Other tangible assets		11,519,970	2,301,426	(2,219)	(4,321,165)	1,477,521	10,975,533
Construction-in-progress		232,705,460	479,820,689	-	-	(403,681,587)	308,844,562
	₩	<u>1,090,959,766</u>	<u>486,548,307</u>	<u>(4,946)</u>	<u>(64,188,480)</u>	<u>17,971</u>	<u>1,513,332,618</u>

(In thousands of won)

		2016					
		Beginning balance	Acquisition	Disposals	Depreciation	Others	Ending balance
Buildings	₩	270,733,945	252,101	-	(7,294,753)	14,287,170	277,978,463
Structures		1,490,651	-	-	(186,977)	5,896,383	7,200,057
Machinery		207,527,245	183,595	-	(44,168,563)	397,907,977	561,450,254
Vehicles		73,292	11,666	-	(82,596)	103,200	105,562
Other tangible assets		18,849,205	1,888,977	(1,033)	(4,051,537)	(5,165,642)	11,519,970
Construction-in-progress		391,581,799	160,656,011	-	-	(319,532,350)	232,705,460
	₩	<u>890,256,137</u>	<u>162,992,350</u>	<u>(1,033)</u>	<u>(55,784,426)</u>	<u>93,496,738</u>	<u>1,090,959,766</u>

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2017 and 2016

13. Property, Plant and Equipment, Continued

(3) The Company recorded construction-in-progress in the amount of ₩377,714,347 thousand and ₩160,250,946 thousand in connection with the construction of buildings and manufacturing facilities for the years ended December 31, 2017 and 2016 respectively.

In addition, ₩10,182,402 thousand and ₩5,921,756 thousand of construction-in-progress has been added from borrowing costs capitalization for the year ended December 31, 2017 and 2016 respectively. Interest rate used to calculate the borrowing costs capitalization is 3.17% for the year ended December 31, 2017.

14. Intangible Assets

(1) Details of intangible assets as of December 31, 2017 and 2016 are as follows:

(In thousands of won)

		December 31, 2017			
		Acquisition cost	Accumulated amortization	Government grants	Carrying amount
Software	₩	27,245,173	(14,742,318)	-	12,502,855
Other intangible assets		1,059,446	(650,249)	-	409,197
Memberships		2,562,188	-	-	2,562,188
Land use rights		69,525,088	(9,270,011)	(60,255,077)	-
Construction-in-progress		3,524,763	-	-	3,524,763
	₩	<u>103,916,658</u>	<u>(24,662,578)</u>	<u>(60,255,077)</u>	<u>18,999,003</u>

(In thousands of won)

		December 31, 2016			
		Acquisition cost	Accumulated amortization	Government grants	Carrying amount
Software	₩	19,864,004	(9,254,120)	-	10,609,884
Other intangible assets		991,967	(479,900)	-	512,067
Memberships		2,062,188	-	-	2,062,188
Land use rights		69,525,088	(7,879,510)	(61,645,578)	-
Construction-in-progress		860,515	-	-	860,515
	₩	<u>93,303,762</u>	<u>(17,613,530)</u>	<u>(61,645,578)</u>	<u>14,044,654</u>

(2) Changes in intangible assets for the years ended December 31, 2017 and 2016 are as follows:

(In thousands of won)

		2017						
		Beginning balance	Acquisition	Disposal	Amortization	Amortization Of government grants	Others	Ending balance
Software	₩	10,609,884	3,658,536	(39,600)	(5,567,397)	-	3,841,432	12,502,855
Other intangible assets		512,067	67,480	-	(170,350)	-	-	409,197
Memberships		2,062,188	500,000	-	-	-	-	2,562,188
Land use rights		-	-	-	(1,390,502)	1,390,502	-	-
Construction-in-progress		860,515	6,523,651	-	-	-	(3,859,403)	3,524,763
	₩	<u>14,044,654</u>	<u>10,749,667</u>	<u>(39,600)</u>	<u>(7,128,249)</u>	<u>1,390,502</u>	<u>(17,971)</u>	<u>18,999,003</u>

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2017 and 2016

14. Intangible Assets, Continued

(2) Changes in intangible assets for the years ended December 31, 2017 and 2016 are as follows, continued:

(In thousands of won)

		2016				
		Beginning balance	Acquisition	Amortization	Amortization of government grants	Ending balance
Software	₩	8,449,124	1,830,427	(4,209,869)	-	10,609,884
Other intangible assets		374,324	71,993	(155,594)	-	512,067
Memberships		2,207,872	511,000	-	-	2,062,188
Land use rights		-	-	(1,390,502)	1,390,502	-
Construction-in-progress		3,254,420	2,765,706	-	-	860,515
	₩	<u>14,285,740</u>	<u>5,179,126</u>	<u>(5,755,965)</u>	<u>1,390,502</u>	<u>14,044,654</u>

(3) The Company performed an impairment test on intangible assets which have indefinite useful lives and no impairment loss recognized for the year ended December 31, 2017 as the recoverable amount exceeded the carrying amount of the intangible assets. The recoverable amount of an intangible assets is the greater of its value in use and its fair value less costs to sell. The value in use is estimated by applying a discount rate that reflects current market assessments of the time value of money and the risks specific to the intangible assets for which estimated future cash flows have not been adjusted, to the estimated future cash flows expected to be generated by the intangible assets.

15. Bonds and Borrowings

(1) Bonds and borrowings as of December 31, 2017 and 2016 are as follows:

(In thousands of won)

		December 31, 2017		December 31, 2016	
		Current	Non-current	Current	Non-current
Bonds	₩	49,926,578	50,000,000	169,917,877	50,000,000
Short-term borrowings		40,000,000	-	-	-
Long-term borrowings		48,750,000	357,235,570	233,776,207	389,363,194
	₩	<u>138,676,578</u>	<u>407,235,570</u>	<u>403,694,084</u>	<u>439,363,194</u>

(2) Details of bonds as of December 31, 2017 and 2016 are as follows:

(In thousands of won)

	Issuance date	Maturity date	Interest rate(%)	December 31, 2017	December 31, 2016
The 1st placement bond	Oct 24, 2014	Oct 24, 2019	3.48	₩ 50,000,000	50,000,000
The 2nd placement bond	Jan 28, 2016	Jan 28, 2017	2.38	-	39,995,381
The 3rd placement bond	Mar 31, 2016	Mar 31, 2017	2.28	-	79,969,875
The 4th placement bond	Aug 12, 2016	Aug 12, 2017	2.00	-	49,952,621
The 5th placement bond	Dec 21, 2017	Dec 21, 2018	2.43	49,926,578	-
				₩ <u>99,926,578</u>	<u>219,917,877</u>

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2017 and 2016

15. Bonds and Borrowings, Continued

(3) Details of short-term borrowings as of December 31, 2017 and 2016 are as follows:

(In thousands of won)

Financial institution	Description	Maturity date	Interest rate (%)	December 31, 2017	December 31, 2016
Korea Development Bank	Loans for equipment	Sep 28, 2018	CD(91D)+0.90 ₩	40,000,000	-
				<u>40,000,000</u>	<u>-</u>

(4) Details of long-term borrowings as of December 31, 2017 and 2016 are as follows:

(In thousands of won)

Financial institution	Description	Maturity date	Interest rate (%)	December 31, 2017	December 31, 2016
Korea Development Bank	Loans for equipment	Aug 21, 2017 Jul 24, 2019 Jun 12, 2020(*1) Dec 15, 2020(*1)	3.79~4.00 ₩ 3.24~3.57 2.86~2.90 2.86	- 50,000,000 50,000,000 40,000,000	50,000,000 50,000,000 50,000,000 40,000,000
Export-Import Bank of Korea	Loans for equipment in foreign currency	Jun 23, 2017	LIBOR+1.90	-	7,251,000
	Overseas investment loans	Jun 14, 2019	LIBOR+1.50	14,463,900	16,314,750
Kookmin Bank	Loans for Equipment	Aug 11, 2020(*1) Sep 28, 2020(*2)	CD(91D)+1.54 CD(91D)+1.43	50,000,000 30,000,000	50,000,000 -
KEB Hana Bank		Nov 14, 2019 (*3) Feb 26, 2021 (*1)	CD(91D)+1.45 CD(91D)+1.60	50,000,000 800,000	100,000,000 800,000
NongHyup Bank		Nov 25, 2019	MOR(1Y)+1.52	50,000,000	50,000,000
Woori Bank		Sep 28, 2020	CD(91D)+1.43	30,000,000	-
Shinhan Bank		Mar 28, 2019 (*3) Sep 28, 2020	3.98 CD(91D)+1.44	- 30,000,000	100,000,000 -
Mizuho Bank, Ltd.	Loans for equipment in foreign currency	Feb 12, 2018 (*3) May 23, 2018 (*3) Jun 26, 2018 (*3) Jul 23, 2018 (*3) Aug 3, 2018 (*3) Nov 28, 2017 (*3) Dec 26, 2017 (*3)	LIBOR+1.35 LIBOR+1.35 LIBOR+1.35 LIBOR+1.35 LIBOR+1.35 LIBOR+1.35 LIBOR+1.35	- - - - - - -	8,879,885 5,359,472 1,672,326 1,557,501 2,685,608 31,898,928 8,371,279
Australia and New Zealand Bank		Apr 28, 2017 Jul 25, 2017 (*3) Sep 22, 2017 (*3)	LIBOR+1.85 LIBOR+1.85 LIBOR+1.85	- - -	11,686,195 12,459,635 12,109,170
Industrial and Commercial Bank of China, Ltd.		Jun 29, 2019	LIBOR+1.50	10,721,670	12,093,652
Current portion of long-term borrowings				(48,750,000)	(233,776,207)
				<u>₩ 357,235,570</u>	<u>389,363,194</u>

(*1) The borrowings have the condition for installment payment every three months starting from two years before their maturities.

(*2) The borrowing can be early redeemed according to the borrowing agreement.

(*3) ₩150,000 million and \$70,330 thousand of borrowings have been early repaid for the year ended December 31, 2017.

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2017 and 2016

15. Bonds and Borrowings, Continued

(5) Changes in liabilities arising from financial activities for the year ended December 31, 2017 are as follows:

		Bonds	Short-term borrowings	Long-term borrowings	Current portion of long-term borrowings	Total
<i>(In thousands of won)</i>						
Balance as of January 1, 2017	₩	219,917,877	-	233,776,206	389,363,193	843,057,276
Fluctuations in financial cash flows:						
Proceeds from borrowings		-	40,000,000	-	90,000,000	130,000,000
Proceeds from issue of bonds		50,000,000	-	-	-	50,000,000
Cost of issue of bonds		(75,520)	-	-	-	(75,520)
Repayment of borrowings		-	-	(236,562,523)	(60,537,587)	(297,100,110)
Repayment of bonds		(170,000,000)	-	-	-	(170,000,000)
		<u>(120,075,520)</u>	<u>40,000,000</u>	<u>(236,562,523)</u>	<u>29,462,413</u>	<u>(287,175,630)</u>
Liquidity transfer		-	-	57,629,885	(57,629,885)	-
Exchange rate effects on foreign currency		-	-	(6,093,568)	(3,960,151)	(10,053,719)
Discount on debentures		84,221	-	-	-	84,221
Balance as of December 31, 2017	₩	<u>99,926,578</u>	<u>40,000,000</u>	<u>48,750,000</u>	<u>357,235,570</u>	<u>545,912,148</u>

(6) The repayment schedule of bonds and borrowings as of December 31, 2017 is as follows:

(In thousands of won)

Period		Bonds	Short-term borrowings	Long-term borrowings	Total
Jan 1, 2017 ~ Dec 31, 2018	₩	50,000,000	40,000,000	48,750,000	138,750,000
Jan 1, 2019 ~ Dec 31, 2019		50,000,000	-	245,485,570	295,485,570
Jan 1, 2020 ~ Dec 31, 2020		-	-	111,650,000	111,650,000
Jan 1, 2020 ~ Dec 31, 2021		-	-	100,000	100,000
	₩	<u>100,000,000</u>	<u>40,000,000</u>	<u>405,985,570</u>	<u>545,985,570</u>

16. Trade and Other Payables

Details of trade and other payables as of December 31, 2017 and 2016 are as follows:

(In thousands of won)

		December 31, 2017		December 31, 2016	
		Current	Non-current	Current	Non-current
Trade payables	₩	14,479,382	-	1,521,209	-
Other payables (*)		115,065,186	-	113,489,469	-
Accrued expenses		28,670,278	2,650,638	23,045,697	2,300,380
	₩	<u>158,214,846</u>	<u>2,650,638</u>	<u>138,056,375</u>	<u>2,300,380</u>

(*) Other payables to related parties as of December 31, 2017 and December 31, 2016 are ₩88,500,736 thousand and ₩96,657,493 thousand, respectively.

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2017 and 2016

17. Emission liability

(1) Emissions rights

The Company has been participating in Emission Trading Scheme in accordance with *the Act on Allocation and Trading of Greenhouse Gas Emission* since 2017. For 2017, the Company has been granted free of charge emission rights of 50,497 tons of greenhouse gas emissions, and there are no emission rights provided as collateral as of December 31, 2017.

(2) Emission liability

Changes in emission liability for the year ended December 31, 2017 are as follows:

(In thousands of won)

		Beginning balance	Increase	Decrease	Ending balance
Emission liability	₩	-	502,080	-	502,080

The Company estimated that it emitted 85,314 tons of carbon dioxide for the year ended December 31, 2017.

18. Other Current Liabilities

Details of other current liabilities as of December 31, 2017 and 2016 are as follows:

(In thousands of won)

		December 31, 2017	December 31, 2016
Advance receipts	₩	53,896,457	58,708,693
Withholdings		2,589,718	1,615,098
	₩	<u>56,486,175</u>	<u>60,323,791</u>

19. Employee Benefits

(1) Recognized liabilities for employee benefits as of December 31, 2017 and December 31 2016 are as follows:

(In thousands of won)

	Explanation	December 31, 2017	December 31, 2016
Defined benefit liabilities	Present value of defined benefit obligations	₩ 33,888,400	23,333,875
	Fair value of plan assets	(29,892,336)	(17,576,610)
		<u>3,996,064</u>	<u>5,757,265</u>
Trade and other payables	Accrued expenses (*)	22,021,288	15,670,445
	Long-term accrued expenses (*)	2,650,638	2,300,380
		<u>24,671,926</u>	<u>17,970,825</u>
		₩ <u>28,667,990</u>	<u>23,728,090</u>

(*) The amounts include the accrued compensation for annual leave, performance bonuses and others payable to employees.

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2017 and 2016

19. Employee Benefits, Continued

(2) Changes in defined benefit liabilities for the years ended December 31, 2017 and 2016 are as follows:

(In thousands of won)

	2017		
	Defined benefit obligations	Plan assets	Net defined benefit liabilities
Beginning balance as of January 1	₩ 23,333,875	(17,576,610)	5,757,265
Included in profit or loss (*)			
Current service cost	8,545,737	-	8,545,737
Interest expense (income)	711,053	(543,512)	167,541
	<u>9,256,790</u>	<u>(543,512)</u>	<u>8,713,278</u>
Remeasurement loss (gain) :			
- Actuarial losses (gains) arising from			
Demographic assumptions	-	-	-
Financial assumptions	(170,311)	-	(170,311)
Experience adjustment	2,696,084	-	2,696,084
- Return on plan assets excluding interest income	-	362,898	362,898
	<u>2,525,773</u>	<u>362,898</u>	<u>2,888,671</u>
Other :			
Contributions to plan assets	-	(13,085,879)	(13,085,879)
Retirement benefit paid	(1,551,339)	1,184,968	(366,371)
Transfer from affiliated company	323,301	(234,201)	89,100
	<u>(1,228,038)</u>	<u>(12,135,112)</u>	<u>(13,363,150)</u>
Ending balance as of December 31	₩ <u>33,888,400</u>	<u>(29,892,336)</u>	<u>3,996,064</u>

(In thousands of won)

	2016		
	Defined benefit obligations	Plan assets	Net defined benefit liabilities
Beginning balance as of January 1	₩ 15,502,179	(6,787,823)	8,714,356
Included in profit or loss :			
Current service cost	6,112,437	-	6,112,437
Interest expense (income)	470,654	(206,082)	264,572
	<u>6,583,091</u>	<u>(206,082)</u>	<u>6,377,009</u>
Remeasurement loss (gain) :			
- Actuarial losses (gains) arising from			
Demographic assumptions	(181,982)	-	(181,982)
Financial assumptions	(135,755)	-	(135,755)
Experience adjustment	2,270,175	-	2,270,175
- Return on plan assets excluding interest income	-	153,008	153,008
	<u>1,952,438</u>	<u>153,008</u>	<u>2,105,446</u>
Other :			
Contributions to plan assets	-	(11,056,300)	(11,056,300)
Retirement benefit paid	(703,833)	320,587	(383,246)
	<u>(703,833)</u>	<u>(10,735,713)</u>	<u>(11,439,546)</u>
Ending balance as of December 31	₩ <u>23,333,875</u>	<u>(17,576,610)</u>	<u>5,757,265</u>

(*) ₩1,075,679 thousand of loss has been added to construction-in-progress for the year ended December 31, 2017

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2017 and 2016

19. Employee Benefits, Continued

(3) Details of plan assets as of December 31, 2017 and 2016 are as follows:

(In thousands of won)

		December 31, 2017	December 31, 2016
Cash and cash equivalents	₩	29,892,336	17,576,610

(4) Defined benefit liabilities

① Principal actuarial assumptions as of December 31, 2017 and 2016 are as follows:

	December 31, 2017	December 31, 2016
Discount rate	3.60%	3.30%
Future salary increases	6.12%	5.92%

In order to calculate present value of defined benefit obligations, the Company determined discount rate based on return of corporate bonds, which are consistent with currency of defined benefit obligations and expected payment term as of the reporting date.

As of December 31, 2017 and 2016, the weighted-average duration of the defined benefit obligation is 7.77 years and 7.53 years respectively.

② As of December 31, 2017, effects on present value of defined benefit obligations if each of significant actuarial assumptions changes within potential reasonable range are as follows (if changed by 1%):

(In thousands of won)

		Salary increases		Discount rate	
		1% Up	1% Down	1% Up	1% Down
Present value of defined benefit obligations	₩	2,967,214	(2,561,357)	(2,505,589)	2,953,427

20. Share Capital and Reserves

(1) Details of share capital as of December 31, 2017 and 2016 are summarized as follows:

(In thousands of won, except par value and number of shares)

		December 31, 2017	December 31, 2016
Number of shares authorized		500,000,000	500,000,000
Par value	₩	2,500	2,500
Number of shares issued		66,165,000	66,165,000
Share capital	₩	165,412,500	165,412,500

(2) Changes in the number of shares issued for the years ended December 31, 2017 and 2016 are as follows:

(In number of shares)

	December 31, 2017		December 31, 2016	
	Number of shares issued	Outstanding stock	Number of shares issued	Outstanding stock
Beginning number of shares	66,165,000	66,165,000	27,568,721	27,568,721
Stock split	-	-	27,568,721	27,568,721
Capital increase	-	-	11,027,558	11,027,558
Ending number of shares	66,165,000	66,165,000	66,165,000	66,165,000

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2017 and 2016

20. Share Capital and Reserves, Continued

(3) Details of share premium as of December 31, 2017 and 2016 are as follows:

(In thousands of won)

	December 31, 2017	December 31, 2016
Additional paid-in capital	₩ 2,496,583,409	2,496,583,409
Irregular capital increase	(9,270,327)	(9,270,327)
	<u>₩ 2,487,313,082</u>	<u>2,487,313,082</u>

(4) Details of accumulated other comprehensive income (loss) as of December 31, 2017 and 2016 are as follows:

(In thousands of won)

	December 31, 2017	December 31, 2016
Defined benefit plan remeasurements	₩ (8,610,119)	(6,420,506)
Equity adjustment arising from investments in associates and joint ventures	4,600,672	11,367,511
	<u>₩ (4,009,447)</u>	<u>4,947,005</u>

(5) Appropriations of retained earnings as of December 31, 2017 and 2016, is as follows:

(In thousands of won)

	2017 (Date of Disposition for 2017 : March 22, 2018)	2016 (Date of Disposition for 2016 : March 24, 2017)
Retained earnings before appropriations	₩ 1,327,734,757	1,424,706,873
Unappropriated retained earnings (undisposed deficit) carried over from prior	1,424,706,873	1,601,530,283
Net income (loss)	(96,972,116)	(176,823,410)
Appropriations of retained earnings	-	-
Unappropriated retained earnings to be carried forward to subsequent year	<u>₩ 1,327,734,757</u>	<u>1,424,706,873</u>

21. Revenue

Details of revenue for the years ended December 31, 2017 and 2016 are as follows:

(In thousands of won)

	2017	2016
Product sales	₩ 375,324,354	240,820,443
Other sales	89,304,980	53,801,578
	<u>₩ 464,629,334</u>	<u>294,622,021</u>

22. Cost of Revenue

Details of cost of revenue for the years ended December 31, 2017 and 2016 are as follows:

(In thousands of won)

	2017	2016
Cost of product sales	₩ 238,530,759	199,420,965
Cost of other sales	94,298,514	68,667,393
	<u>₩ 332,829,273</u>	<u>268,088,358</u>

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2017 and 2016

23. Selling, General and Administrative Expenses

Details of selling, general and administrative expenses for years ended December 31, 2017 and 2016 are as follows:

(In thousands of won)

	2017	2016
Salaries	₩ 12,791,255	11,120,503
Bonuses	7,282,797	5,005,847
Retirement benefits	1,267,966	1,087,455
Employee welfare	5,849,421	4,322,572
Depreciation	6,486,402	4,246,229
Amortization of intangible assets	3,008,710	1,829,666
Repairs and maintains	377,432	1,008,278
Supplies expenses	581,324	832,904
Utility expenses	213,688	207,601
Travel expenses	1,350,522	1,441,789
Service fee	11,514,320	11,901,152
Training expenses	5,228,274	5,498,193
Electronic processing expenses	995,820	878,147
Rent expense	236,744	235,209
Taxes and dues	997,366	1,200,086
Others	7,638,618	6,140,671
	₩ <u>65,820,659</u>	<u>56,956,302</u>

24. Nature of Expenses

Details of expenses classified by nature for the years ended December 31, 2017 and 2016 are as follows:

(In thousands of won)

	2017	2016
Changes in inventories	₩ (17,154,371)	(21,633,321)
Raw materials used	75,684,772	24,934,106
Employee benefits	148,689,762	110,259,584
Depreciation and amortization	69,926,227	60,149,889
Supplies expenses	33,346,233	57,984,312
Utility expenses	17,716,247	14,699,093
Service fees	22,787,981	27,928,209
Others	47,653,081	50,722,788
	₩ <u>398,649,932</u>	<u>325,044,660</u>

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2017 and 2016

25. Other Income and Expenses

Details of other income and expenses for the years ended December 31, 2017 and 2016 are as follows:

(In thousands of won)

	2017	2016
Other income		
Rental revenues	₩ 487,683	562,290
Fees revenues	131,759	205,488
Miscellaneous gains	270,541	450,052
	₩ <u>889,983</u>	<u>1,217,830</u>
Other expenses		
Donations	₩ 19,900	24,343
Loss on disposal of property, plant and equipment	4,946	1,033
Loss on disposal of intangible assets	39,600	-
Miscellaneous losses	71,797	13,634
	₩ <u>136,234</u>	<u>39,010</u>

26. Finance Income and Costs

Details of finance income and costs for the years ended December 31, 2017 and 2016 are as follows:

(In thousands of won)

	2017	2016
Finance income		
Interest income	₩ 9,949,872	3,639,650
Gain on foreign currency translations	3,512,447	2,825,232
Gain on foreign currency transactions	9,757,456	7,883,685
	₩ <u>23,219,775</u>	<u>14,348,567</u>
Finance costs		
Interest expenses	₩ 5,683,983	19,254,372
Loss on foreign currency translations	331,259	3,962,749
Loss on foreign currency transactions	11,238,606	9,534,066
Loss on valuation of derivatives	59,091,484	54,031,894
	₩ <u>76,345,332</u>	<u>86,783,081</u>

27. Income Tax Expense (Benefit)

(1) Income tax expense (benefit) for the years ended December 31, 2017 and 2016 is as follows:

(In thousands of won)

	2017	2016
Current income tax expense:		
Current income tax expense	₩ 1,865,764	-
Deferred tax expense:		
Temporary differences	(21,354,723)	(54,903,810)
Income taxes recognized directly in equity	466,322	826,237
Income tax benefit	₩ <u>(19,022,637)</u>	<u>(54,077,573)</u>

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2017 and 2016

27. Income Tax Expense (Benefit), Continued

(2) The relationship between tax expense (benefit) and accounting profit (loss) for the year ended December 31, 2017 and 2016 is as follows:

(In thousands of won)

	2017	2016
Loss before income tax	₩ (115,994,753)	(230,900,983)
Income taxes at statutory tax rate	(28,070,730)	(55,878,038)
Adjustments:		
Non-deductible expense	151,764	851,359
Changes in unrecognized deferred tax assets	10,177,594	949,106
Others	(1,281,265)	-
Income tax benefit	₩ (19,022,637)	(54,077,573)

(3) Income taxes recognized directly in equity for the year ended December 31, 2017 and 2016 are as follows:

(In thousands of won)

	2017			2016		
	Before tax	Tax expense	Net of tax	Before tax	Tax expense	Net of tax
Defined benefit plan remeasurements	₩ (2,888,671)	699,058	(2,189,613)	(2,105,448)	509,519	(1,595,929)
Equity adjustments arising from investments in associates and joint ventures	(6,534,103)	(232,736)	(6,766,839)	1,386,549	316,718	1,703,267
	₩ (9,422,774)	466,322	(8,956,452)	(718,899)	826,237	107,338

(4) Changes in temporary differences and deferred tax assets (liabilities) as of December 31, 2017 and 2016, are as follows:

(In thousands of won)

	2017			
	January 1, 2017	Profit or loss	Other comprehensive income(loss)	December 31, 2017
Accrued expenses	₩ 2,132,868	553,031	-	2,685,899
Net defined benefit obligations	1,154,753	(903,854)	699,058	949,957
Long-term accrued expenses	556,692	84,762	-	641,454
Loss (gain) on foreign currency translations	2,613,108	(3,192,310)	-	(579,202)
Provision for loss on valuation of finished goods	2,035,998	92,060	-	2,128,058
Provision for loss on valuation of work in progress	587,030	(585,392)	-	1,638
Depreciation	52,408	109,534	-	161,942
Provision for performance bonus	2,768,614	1,093,882	-	3,862,496
Derivative liabilities	453,621,968	14,300,139	-	467,922,107
Equity-accounted investees	(1,052,926,863)	13,383,015	(232,736)	(1,039,776,584)
Accrued income	(645,778)	630,332	-	(15,446)
Emission liability	-	121,503	-	121,503
Tax loss carryforward	45,877,938	(12,607,373)	-	33,270,565
Tax credit carryforward	15,500,650	7,809,071	-	23,309,721
Deferred tax liabilities (assets)	₩ (526,670,614)	20,888,440	466,322	(505,315,892)

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2017 and 2016

27. Income Tax Expense (Benefit), Continued

(4) Changes in temporary differences and deferred tax assets (liabilities) as of December 31, 2017 and 2016, are as follows, continued:

(In thousands of won)

		2016		
		January 1, 2016	Profit or loss	Other comprehensive income(loss) December 31, 2016
Accrued expenses	₩	2,163,027	(30,159)	-
Net defined benefit obligations		1,777,623	(1,132,389)	509,519
Long-term accrued expenses		462,117	94,575	-
Loss (gain) on foreign currency translations		1,321,945	1,291,163	-
Provision for loss on valuation of finished goods		2,727,725	(691,726)	-
Provision for loss on valuation of work in progress		203,948	383,082	-
Depreciation		-	52,408	-
Provision for performance bonus		1,481,146	1,287,468	-
Derivative liabilities		440,546,250	13,075,718	-
Equity-accounted investees		(1,099,553,749)	46,310,168	316,718
Accrued income		-	(645,779)	-
Tax loss carryforward		35,155,239	10,722,699	-
Tax credit carryforward		32,140,305	(16,639,655)	-
Deferred tax liabilities (assets)	₩	<u>(581,574,424)</u>	<u>54,077,573</u>	<u>826,237</u>
				<u>(526,670,614)</u>

(5) As of December 31, 2017 and December 31, 2016, the Company did not recognize ₩104,277,416 thousand and ₩89,690,264 thousand of deferred tax assets, respectively, with respect to investments in associates and joint ventures, tax credit carryforward, and so forth as the Company determines that it is not probable that they will be realized in foreseeable future.

28. Loss per Share

(1) Basic loss per share for the years ended December 31, 2017 and 2016 is calculated as follows:

(In won)

	2017	2016
Net loss for the year	₩ (96,972,115,811)	(176,823,409,527)
Weighted-average number of ordinary shares outstanding	66,165,000	56,764,459
Basic loss per share	₩ <u>(1,466)</u>	<u>(3,115)</u>

(2) Weighted-average number of ordinary shares outstanding

	2017	2016
Issued ordinary shares at the beginning	₩ 66,165,000	27,568,721
Effect of stock split	-	27,568,721
Capital increase	-	1,627,017
Weighted-average number of ordinary shares as of December 31, 2017	₩ <u>66,165,000</u>	<u>56,764,459</u>

(3) The Company has no dilutive potential shares; therefore, diluted loss per share is equal to the basic loss per share.

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2017 and 2016

29. Commitments and Contingencies

(1) Biogen Therapeutics Inc. has an option to purchase additional shares of Samsung Bioepis Co., Ltd. that would allow them to increase their ownership percentage up to 49.9%. The option price is the sum of principal amount and accumulated interest. The value of the financial derivative liabilities may be changed significantly depending on the market situation and the exercise of the option. If Biogen Therapeutics Inc. does not exercise option until the expiration (in 2018), the judgment of control could be changed (See Notes 6 and 12).

(2) As of December 31, 2017, details of the Company's borrowings commitment with financial institutions are summarized as follows:

(In thousands of won, USD)

Financial institution	Remarks	Currency	Limit	Outstanding balance
Woori Bank	Bank overdraft	KRW	10,000,000	-
	B2B PLUS	KRW	12,000,000	-
	Loans for equipment	KRW	70,000,000	30,000,000
KEB Hana Bank	Bank overdraft	KRW	10,000,000	-
	Loans for equipment	KRW	50,000,000	50,000,000
	Loans for equipment	KRW	50,000,000	800,000
	Comprehensive limit	KRW	40,000,000	-
Shinhan Bank	General loan	KRW	19,214,000	-
	Bank overdraft	KRW	10,000,000	-
	Loans for equipment	KRW	70,000,000	30,000,000
Kookmin Bank	Loans for equipment	KRW	50,000,000	50,000,000
	Loans for equipment	KRW	60,000,000	30,000,000
NongHyup Bank	Loans for equipment	KRW	50,000,000	50,000,000
Korea Development Bank	Loans for equipment	KRW	50,000,000	50,000,000
	Loans for equipment	KRW	50,000,000	50,000,000
	Loans for equipment	KRW	40,000,000	40,000,000
	Loans for equipment	KRW	40,000,000	40,000,000
	Loans for equipment	KRW	40,000,000	40,000,000
Export-Import Bank of Korea	Overseas investment loans	USD	13,500	13,500
Industrial and Commercial Bank of China, Ltd.	Loans for equipment in foreign currency	USD	50,000	10,007

(3) As of December 31, 2017, details of guarantees provided by financial institutions are summarized as follows:

(In thousands of won, USD)

Financial institution	Remarks	Currency	Limit	Currency	Outstanding balance
Woori Bank	Payment guarantees	KRW	9,500,000	KRW	2,071,579
	Letter of credit	USD	8,500	USD	-
	Mortgage-backed payment guarantee	USD	17,800	USD	-
Shinhan Bank	Letter of credit	USD	10,000	USD	-
	Foreign currency payment guarantee	USD	10,000	USD	-
Kookmin Bank	Foreign currency payment guarantee	KRW	30,000,000	USD	-

29. Commitments and Contingencies, Continued

(4) In connection with a lease agreement with Incheon Metropolitan city ("Incheon") related to the use of land, the Company is exempt from the full amount of rent because the Company meets the requirements for exemption of rent in accordance with the *Common Property Management Ordinance of Incheon*. The Company recognized an intangible asset at the fair value of the rent that is exempt from Incheon and government grant as a reduction of assets at the same value. The period of land use without payment and exemption requirement of the full amount of rent are as follows:

(i) Period

A lease term is for a total of 50 years. The initial lease period is for 20 years from the land lease contract date, April 28, 2011, and the Company can renew the contract for another 30 years in successive 10 year increments.

(ii) Exemption requirements

The Company whose foreign investment amounts to more than USD 20 million for 5 years since the land lease contract date is subject to the exemption from *Common Property Management Ordinance of Incheon*. Afterwards, the Company, including foreign invested enterprises that meet the average daily employee count of above 300, is subject to exemption.

30. Related Parties

(1) As of December 31, 2017, Samsung C&T Corporation is the parent of the Company, and the percentage of ownership it has over the Company is 43.44%.

(2) Details of related parties as of December 31, 2017 are as follows:

Relationship	Companies
Parent	Samsung C&T Corporation
Entity which has significant influence	Samsung Electronics Co., Ltd.
Associates and Joint ventures	Samsung Bioepis Co., Ltd. Archigen Biotech Limited
Other	Samsung Welstory Inc., Samsung C&T Deutschland GmbH, Samsung Research America, Inc, Samsung Electronics Sales Co., Ltd.
The Enterprise Group (*)	Samsung Card Co., Ltd., Samsung Engineering Co., Ltd., S-1 Corporation, Samsung SDS Co., Ltd., Samsung Economic Research Institute, Multicampus Corporation, Samsung Fire & Marine Insurance Co., Ltd., Samsung Life Insurance Co., Ltd., Hotel Shilla Co., Ltd., Cheil Worldwide Inc, Samsung Securities Co., Ltd.

(*) The entities are not included in the scope of related parties in accordance with K-IFRS No.1024 but are included in the scope of Enterprise Group designated by Fair Trade Commission of the Republic of Korea.

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2017 and 2016

30. Related Parties, Continued

(3) Account balances with related parties as of December 31, 2017 and 2016 are as follows:

(In thousands of won)

Relationship	Companies	December 31, 2017				
		Receivables			Payables	
		Trade receivables	Other receivables	Accrued income	Other payables	Advance receipts
Entity which has Significant influence	Samsung Electronics Co., Ltd.	-	-	-	22,328	-
Associates and Joint ventures	Samsung Bioepis Co., Ltd.(*1)	2,503,684	647,378	52,134	-	4,463,334
	Archigen Biotech Limited	-	514,990	527,471	5,161	-
Other	Samsung Welstory Inc.	-	3,138	-	682,238	-
	Samsung C&T Deutschland GmbH	-	-	-	45,015	-
	Samsung Research America, Inc	-	-	-	193,902	-
The Enterprise Group	Samsung Card Co., Ltd.	-	-	-	64,884,955	-
	Samsung Engineering Co., Ltd.	-	-	-	16,759,196	-
	S-1 Corporation	-	-	-	3,192,538	-
	Samsung SDS Co., Ltd.	-	-	-	2,333,707	-
	Samsung Economic Research Institute	-	-	-	298,081	-
	Samsung Fire & Marine Insurance Co., Ltd.	-	-	-	2,177	-
	Others(*2)	-	-	-	81,438	-
		<u>₩ 2,503,684</u>	<u>1,165,506</u>	<u>579,605</u>	<u>88,500,736</u>	<u>4,463,334</u>

(*1) The amount of trade receivables reflects offset against advance receipts of ₩ 65,027 thousand of Samsung Bioepis Co., Ltd.

(*2) Transactions with Multicampus Corporation, Hotel Shilla Co.,Ltd. and Cheil Worldwide Inc. were included.

(In thousands of won)

Relationship	Companies	December 31, 2016		
		Receivables		Payables
		Other receivables	Accrued income	Other payables
Parent	Samsung C&T Corporation	₩ -	-	13,484
Entity which has significant influence	Samsung Electronics Co., Ltd.	-	-	18,270
Associates and Joint ventures	Samsung Bioepis Co., Ltd.	-	78,972	-
	Archigen Biotech Limited	-	10,632	19,678
Other	Samsung Welstory Inc.	-	10,604	512,524
	Samsung C&T Deutschland GmbH	-	-	74,659
The Enterprise Group	Samsung Card Co., Ltd.	-	-	67,292,579
	Samsung Engineering Co., Ltd.	-	-	26,901,600
	S-1 Corporation	-	-	1,058,094
	Samsung SDS Co., Ltd.	-	-	674,323
	Samsung Economic Research Institute	-	-	98,182
	Samsung Fire & Marine Insurance Co., Ltd.	-	-	432
	Others(*)	-	-	13,346
		<u>₩ 100,208</u>	<u>19,678</u>	<u>96,657,493</u>

(*) Transactions with Multicampus Corporation and Hotel Shilla Co., Ltd., were included.

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2017 and 2016

30. Related Parties, Continued

(4) Significant transactions which occurred in the normal course of business with related parties for the years ended December 31, 2017 and 2016 are summarized as follows:

(In thousands of won)

Relationship	Companies	2017				
		Revenue		Expense		Asset acquisition
		Revenue	Others	Service fee	Others	
Parent	Samsung C&T Corporation	₩ -	-	143,632	16,992	952,253
Entity which has significant influence	Samsung Electronics Co., Ltd.	-	-	-	27,640	-
Associates and Joint ventures	Samsung Bioepis Co., Ltd.	18,423,815	542,150	-	-	-
	Archigen Biotech Limited	-	23,722	-	9,384	-
Others	Samsung Welstory Inc.	-	30,057	-	6,710,571	-
	Samsung C&T Deutschland GmbH	-	-	60,479	248,009	-
	Samsung Research America, Inc.	-	-	2,448,390	-	-
	Samsung Electronics Sales Co., Ltd.	-	3,590	-	13,160	-
The Enterprise Group	Samsung Card Co., Ltd.	-	-	-	207,225	-
	Samsung Engineering Co., Ltd.	-	-	-	-	310,898,297
	S-1 Corporation	-	-	6,356,280	528,938	4,283,760
	Samsung SDS Co., Ltd.	-	-	-	10,815,639	6,240,750
	Samsung Economic Research Institute	-	-	1,075,363	1,273,498	-
	Samsung Fire & Marine Insurance Co., Ltd.(*1)	-	297,954	2,357	3,288,498	-
	Others(*2)	-	4,630	119,046	1,405,954	-
		₩ 18,423,815	902,103	10,205,547	24,545,508	322,375,060

(*1) Other revenue is actual interest income from plan assets, which is recognized as deduction of retirement benefits.

(*2) Transactions with Multicampus Corporation, Hotel Shilla Co.,Ltd., Cheil Worldwide Inc., and Samsung Life Insurance Co., Ltd. were included.

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2017 and 2016

30. Related Parties, Continued

(4) Significant transactions which occurred in the normal course of business with related parties for the years ended December 31, 2017 and 2016 are summarized as follows, continued:

(In thousands of won)

Relationship	Companies	2016				
		Revenue		Expense		Asset acquisition
		Other revenue	Others	Service fee	Others	
Parent	Samsung C&T Corporation	₩ -	-	143,846	23,035	-
Entity which has significant influence	Samsung Electronics Co., Ltd.	-	-	-	18,059	-
Associates and Joint Ventures	Samsung Bioepis Co., Ltd.	205,488	452,406	-	-	-
	Archigen Biotech Limited	-	329,879	-	-	-
Others	Samsung Welstory Inc.	-	37,686	4,196	4,978,265	-
	Samsung C&T Deutschland GmbH	-	-	367,481	15,503	-
The Enterprise Group	Samsung Card Co., Ltd.	-	-	-	256,607	-
	Samsung Engineering Co., Ltd.	-	-	-	-	161,183,000
	S-1 Corporation	-	-	5,242,071	392,710	556,150
	Samsung SDS Co., Ltd.	-	-	48,000	7,742,700	1,778,750
	Samsung Economic Research Institute	-	-	197,271	1,293,383	-
	Samsung Fire & Marine Insurance Co., Ltd.(*1)	-	130,664	2,089	3,494,792	-
	Samsung Securities Co., Ltd.	-	-	-	2,699,539	-
	Others(*2)	-	2,640	55,416	525,582	14,269
		₩ 205,488	953,275	6,060,370	21,440,175	163,532,169

(*1) Other revenue is actual interest income from plan assets, which is recognized as deduction of retirement benefits.

(*2) Transactions with Multicampus Corporation, Hotel Shilla Co.,Ltd., Cheil Worldwide Inc. and Samsung Life Insurance Co., Ltd. were included.

(5) Key management personnel compensation for each of the following categories for the years ended December 31, 2017 and 2016 are summarized as follows:

(In thousands of won)

	2017	2016
Short-term employee benefits	₩ 2,314,139	2,200,739
Long-term employee benefits	1,372,888	1,094,035
Retirement benefits	248,155	463,960
	₩ 3,935,182	3,758,734

Key management refers to the registered directors and audit committee. Compensation of the Company's key management personnel includes salaries, non-cash benefits, and contributions to a post-employment defined benefit plan. In addition, the Company has granted long-term incentives to the Company's key management personnel based on its three-year management performance. The estimated amount of liabilities arising from such incentives are recognized as accrued expenses and long-term accrued expenses.

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2017 and 2016

30. Related Parties, Continued

(6) The Company made a lease agreement with Incheon Metropolitan city ("Incheon") related to the use of land and subleased the land to Samsung Bioepis Co., Ltd., its associate, in consent of Incheon.

(7) Details of financial transactions with related parties for the years ended December 31, 2017 and 2016 are as follows:

(In thousands of won)

	Companies	2017	2016	Description
Associates and Joint Ventures	Samsung Bioepis Co., Ltd. ₩	200,000,000	200,000,000	Equity contribution in cash
	Archigen Biotech Limited	-	34,653,000	Equity contribution in cash

31. Statements of Cash Flows

Significant transactions not involving cash flows of investing and financing activities for the years ended December 31, 2017 and 2016 are summarized as follows:

(In thousands of won)

	2017	2016
Increase in other payables due to acquisition of property, plant and equipment ₩	18,761,648	86,508,215
Transfer construction-in-progress to property, plant and equipment	407,540,989	420,024,932

32. Financial Risk Management

The Company is exposed to credit risk, liquidity risk, and market risk in connection with its financial instruments. This note presents information about the Company's exposure to each of the above risks, the Company's objectives, policies and processes for measuring and managing risks, and the Company's management of capital. Further quantitative disclosures are included throughout these financial statements.

(1) Financial risk management

The Company's risk management policies are established to identify and analyze the risks faced by the Company, to set appropriate risk limits and controls, and to monitor risks and adherence to limits. Risk management policies and systems are reviewed regularly to reflect changes in market conditions and the Company's activities. The Company, through its training and management standards and procedures, aims to maintain a disciplined and constructive control environment in which all employees understand their roles and obligations. The Board of Directors oversees how management monitors compliance with the Company's risk management policies and procedures and reviews the adequacy of the risk management framework in relation to the risks faced by the Company.

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2017 and 2016

32. Financial Risk Management, Continued

(2) Credit risk

Credit risk is the risk of financial loss to the Company if a customer or counterparty to a financial instrument fails to meet its contractual obligations and arises principally from the Company's receivables from customers.

The carrying amount of financial assets represents the maximum credit exposure. The maximum exposure to credit risk as of December 31, 2017 and 2016 are as follows:

(In thousands of won)

		December 31, 2017	December 31, 2016
Cash and cash equivalents	₩	358,088,091	230,337,601
Short-term financial instruments		-	970,000,000
Long-term financial instruments		7,000	8,000
Trade and other receivables		25,817,150	75,630,820
Deposits provided		43,165,000	37,764,300
	₩	<u>427,077,241</u>	<u>1,313,740,721</u>

(3) Liquidity risk

Liquidity risk is the risk that the Company will encounter difficulty in meeting the obligations associated with its financial liabilities that are settled by delivering cash or other financial assets. The Company's approach to managing liquidity is to ensure, as far as possible, that it will always have sufficient liquidity to meet its liabilities when they are due, under both normal and stressed conditions, without incurring unacceptable losses or risking damage to the Company's reputation.

The Company monitors its cash flows through long-term and short-term management strategies and ensures it has sufficient cash on demand to meet expected operational expenditures. This excludes the potential impact of extreme circumstances that cannot reasonably be predicted.

The Company establishes short-term and long-term cash management plans to manage liquidity risk. The Company matches maturity structures of financial assets and liabilities through analyzing and reviewing cash flow budget and actual cash flow. Management believes that the Company is able to redeem its financial liabilities through operating cash flows and cash inflows of financial assets.

The following is the contractual maturities of financial liabilities as of December 31, 2017 and 2016, including estimated interest payments and excluding the impact of netting agreements. The Company does not expect that the cash flows included in the maturity analysis could occur significantly earlier or at different amounts.

(In thousands of won)

		December 31, 2017				
		Carrying amount	Contractual cash flows	Less than 1 year	1-2 year	2-3 year
Trade and other payables	₩	160,865,484	160,865,484	158,214,846	2,650,638	-
Bonds		99,926,578	104,407,787	52,993,172	51,414,615	-
Short-term borrowings		40,000,000	40,760,285	40,760,285	-	-
Long-term borrowings		<u>405,985,570</u>	<u>428,612,072</u>	<u>60,718,045</u>	<u>254,099,571</u>	<u>113,693,973</u>
	₩	<u>706,777,632</u>	<u>734,645,628</u>	<u>312,686,348</u>	<u>308,164,824</u>	<u>113,693,973</u>

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2017 and 2016

32. Financial Risk Management, Continued

(3) Liquidity risk, continued

(In thousands of won)

		December 31, 2016				
		Carrying amount	Contractual cash flows	Less than 1 year	1-2 year	More than 2-3 year 3 years
Trade and other payables ₩	140,356,755	140,356,755	138,056,375	2,300,380	-	-
Bonds	219,917,877	226,028,338	172,875,223	1,738,500	51,414,615	-
Long-term borrowings	623,139,402	657,162,045	248,004,881	50,000,275	306,684,036	52,472,853
₩	<u>983,414,034</u>	<u>1,023,547,138</u>	<u>558,936,479</u>	<u>54,039,155</u>	<u>358,098,651</u>	<u>52,472,853</u>

(4) Currency risk

The Company is exposed to currency risk on sales, purchases, and borrowings. The currencies in which these transactions primarily are denominated are USD and so forth. The Company establishes a plan to manage currency risk through currency forwards and so forth, if considered necessary.

(i) Exposure to currency risk

The Company's exposure to foreign currency risk as of December 31, 2017 and 2016 is as follows:

(In thousands of won)

		December 31, 2017				December 31, 2016		
		USD	EUR	CHF	JPY	USD	EUR	CHF
Assets:								
Cash equivalents ₩	21,256,612	-	-	-	-	-	-	-
Trade and other receivables	20,480,232	14,072	-	-	-	64,553,873	22,687	-
Liabilities:								
Borrowings	(25,185,570)	-	-	-	-	(132,339,402)	-	-
Trade and other payables	(10,869,294)	(342,467)	(36,421)	(3,796)	(1,649,147)	(521,643)	(1,469)	(1,469)
	<u>5,681,980</u>	<u>(328,395)</u>	<u>(36,421)</u>	<u>(3,796)</u>	<u>(69,434,676)</u>	<u>(498,956)</u>	<u>(1,469)</u>	<u>(1,469)</u>

(ii) The following significant exchange rates applied for the years ended December 31, 2017 and 2016 is as follow:

(In won)

		Average rate		Reporting date spot rate	
		2017	2016	December 31, 2017	December 31, 2016
USD ₩		1,130.84	1,160.50	1,071.40	1,208.50
EUR		1,276.39	1,283.30	1,279.25	1,267.60
CHF		1,148.72	1,177.36	1,094.72	1,181.33
JPY		1,008.52	1,068.17	949.11	1,036.81

Samsung Biologics Co., Ltd.
Notes to the Financial Statements
For the years ended December 31, 2017 and 2016

32. Financial Risk Management, Continued

(4) Currency risk, continued

(iii) Sensitivity analysis

A strengthening (weakening) of won, a change of 10% in foreign currency exchange rate as indicated below, against EUR, CHF, and USD as of December 31, 2017 and 2016 would have increased (decreased) profit or loss (before tax) by the amounts shown below. This analysis is based on foreign currency exchange rate variances that the Company considered to be reasonably possible at the reporting date.

(In thousands of won)

		December 31, 2017		December 31, 2016	
		10% increase	10% decrease	10% increase	10% decrease
USD	₩	568,198	(568,198)	(6,943,468)	6,943,468
EUR		(32,840)	32,840	(49,896)	49,896
CHF		(3,642)	3,642	(147)	147
JPY		(380)	380	-	-
	₩	<u>531,336</u>	<u>(531,336)</u>	<u>(6,993,511)</u>	<u>6,993,511</u>

(5) Interest rate risk

The Company is exposed to interest rate risk because they raised funds with interests paid at variable rate. The effect of changes in interest rates by 0.5% applied to the profit and loss (before tax) as of December 31, 2017 and 2016 is as follows:

(In thousands of won)

		December 31, 2017		December 31, 2016	
		0.5% increase	0.5% decrease	0.5% increase	0.5% decrease
Interest expense	₩	(1,047,325)	1,047,325	(1,602,327)	1,602,327

(6) Capital management

The Company's objectives when managing capital are to safeguard the Company's ability to continue as a going concern in order to provide returns for shareholders and benefits for other stakeholders and to maintain an optimal capital structure to reduce the cost of capital.

The debt-to-equity ratio and net borrowing ratio as of December 31, 2017 and 2016 are as follows:

(In thousands of won)

		December 31, 2017	December 31, 2016
Total liabilities (A)	₩	3,206,640,271	3,450,636,647
Total equity (B)		3,976,450,892	4,082,379,460
Cash and cash equivalents (C)		358,088,091	230,337,601
Total bonds and borrowings (D)		545,912,148	843,057,279
Debt-to-equity ratio (A/B)		80.64%	84.53%
Adjusted net borrowing ratio ((D-C)/B)		4.72%	15.01%